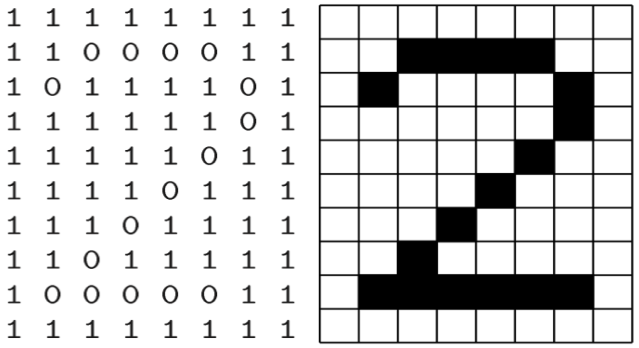
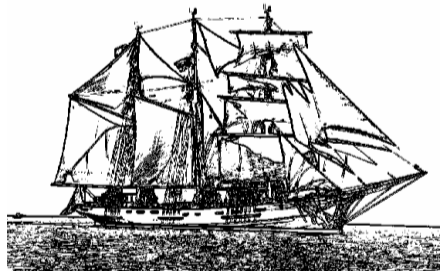
MAP201 : Découverte des Mathématiques Appliquées

Images

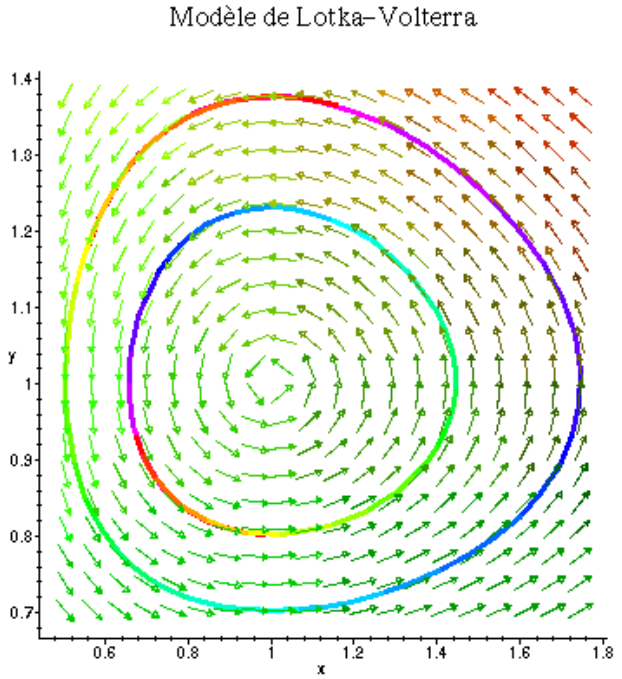






Jérome Lesaint

Equa Diff

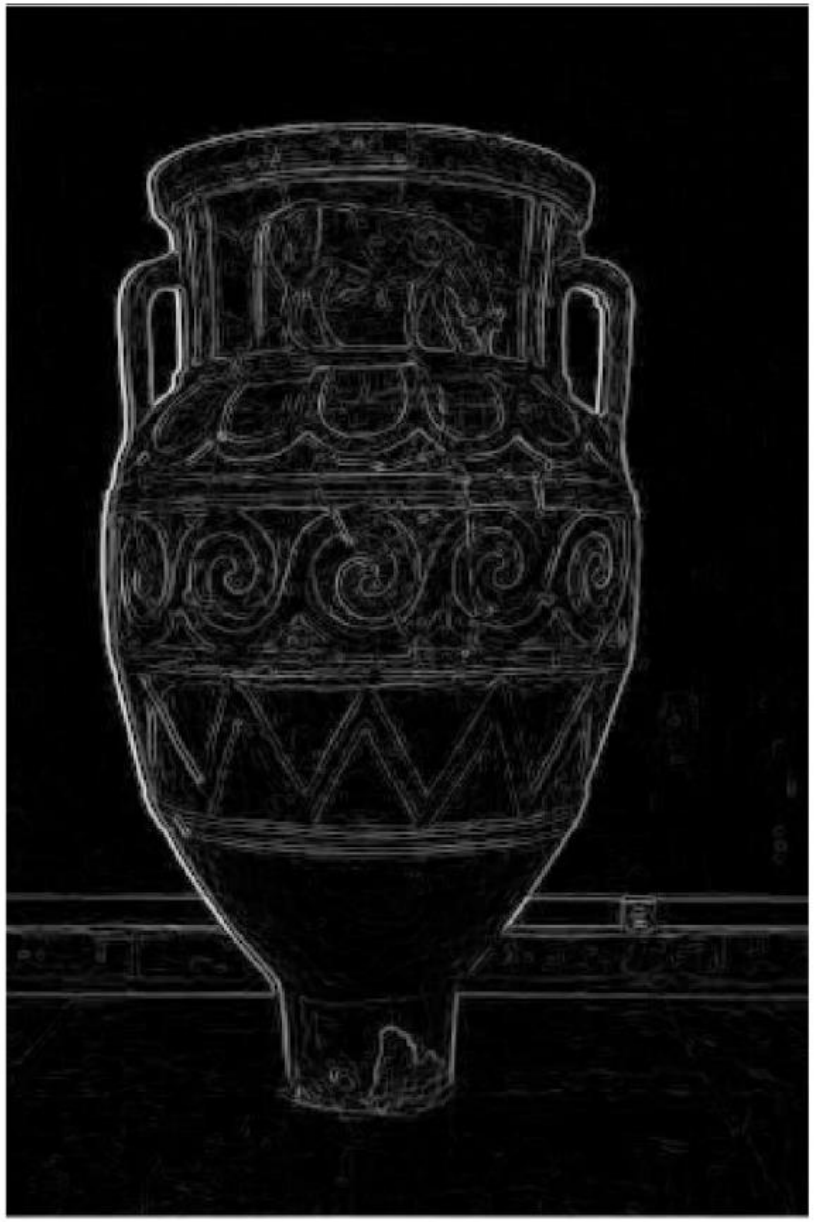


Mickael Nahon

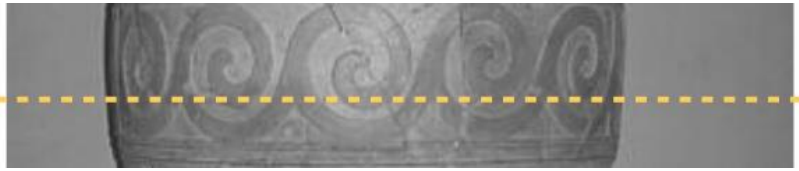
4ÈME COURS :
FILTRES DÉRIVATEURS, CONTOURS

Comment trouver le(s) contour(s) de l'amphore ?

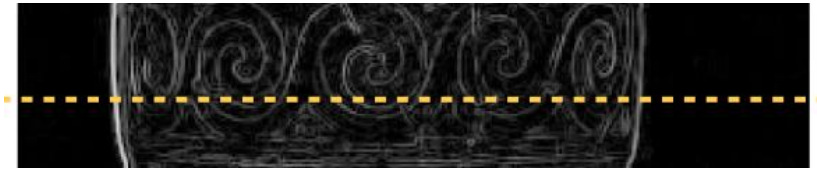




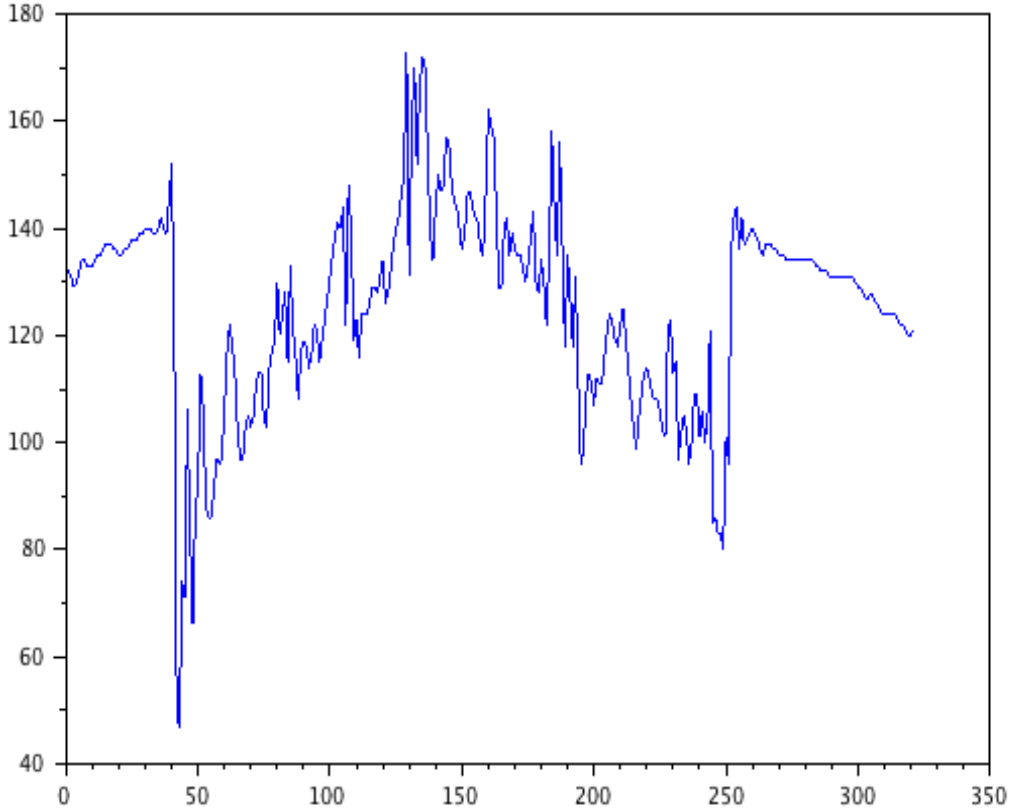
Norme du gradient obtenu par différences finies

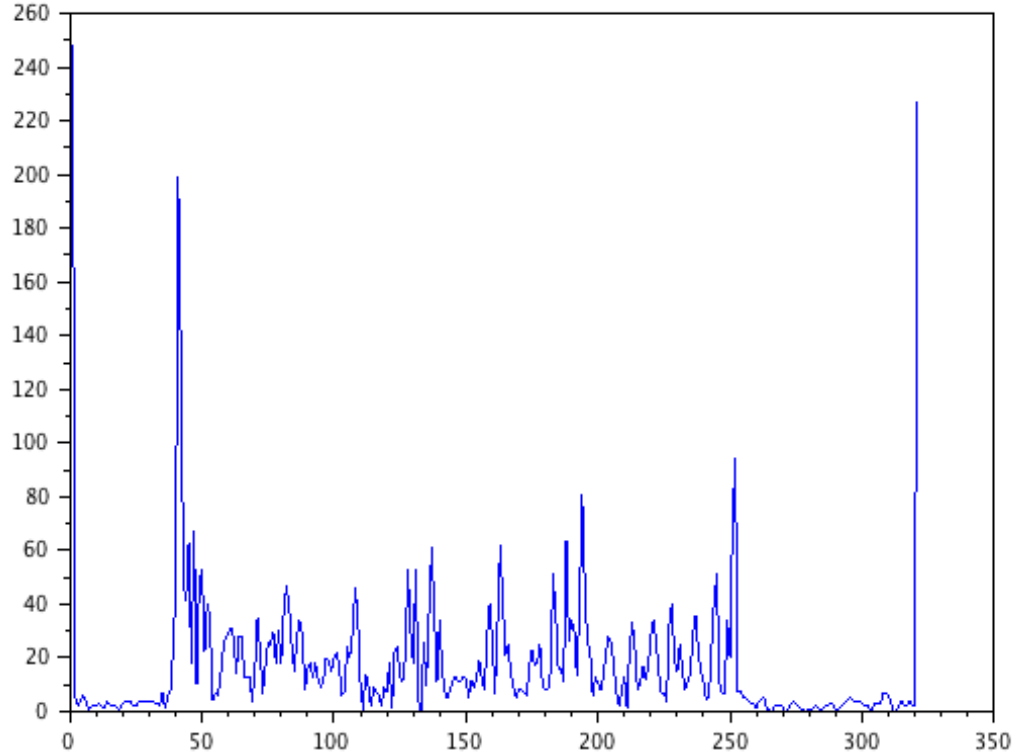


Coupe de l'image



Norme du gradient



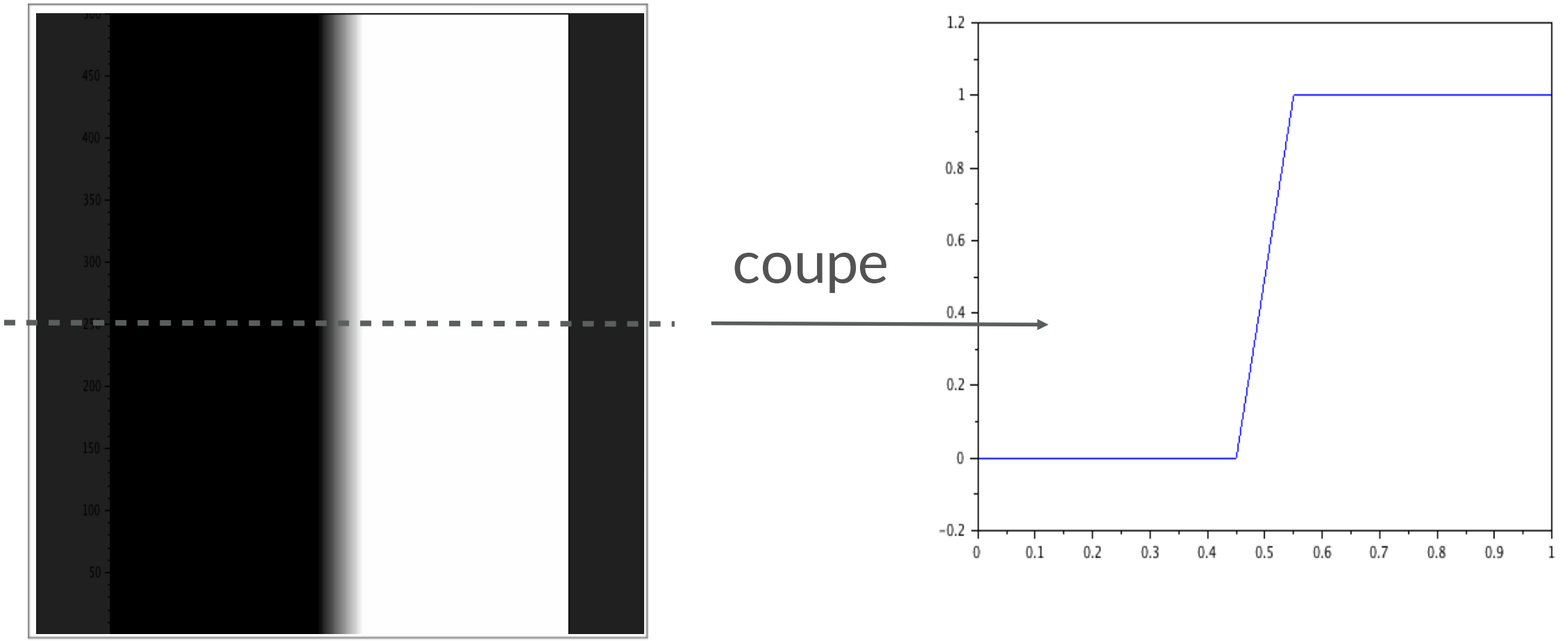


Plan

- Vers la détection de contours
  - besoin de "dériver une image"
  - Fonctions de plusieurs variables
  - dérivabilité, gradient
- Application aux images discrètes
- Détection de contours

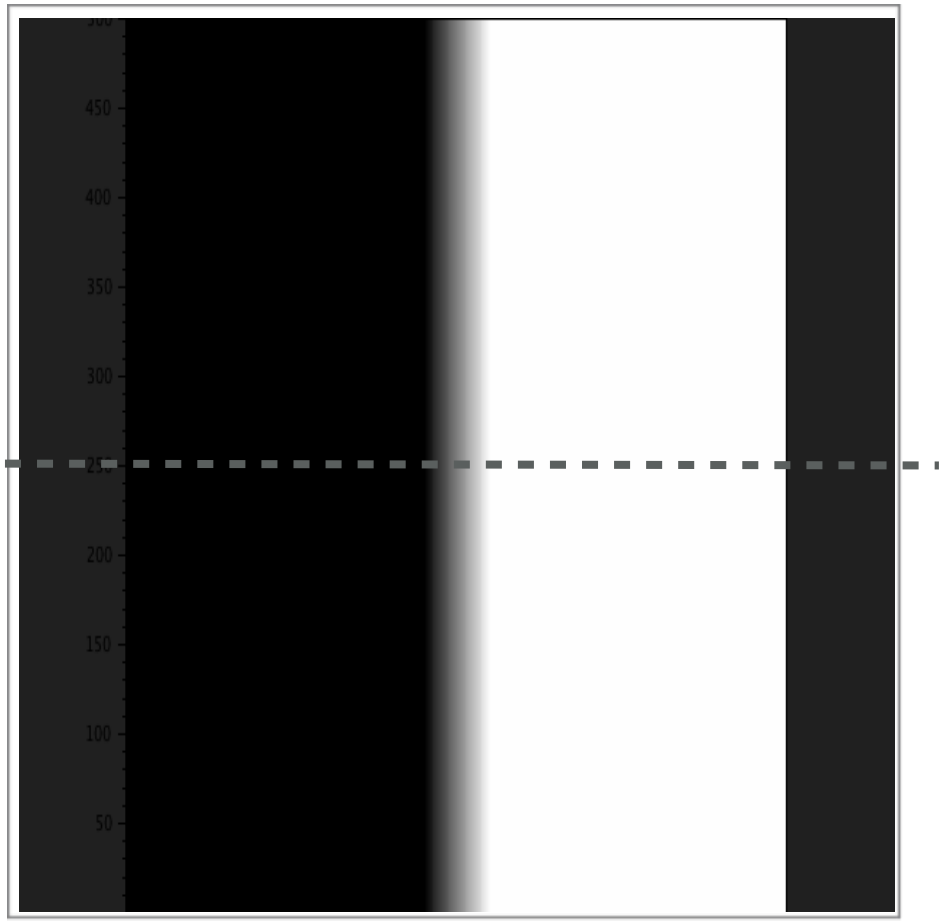
Vers la détection de contours

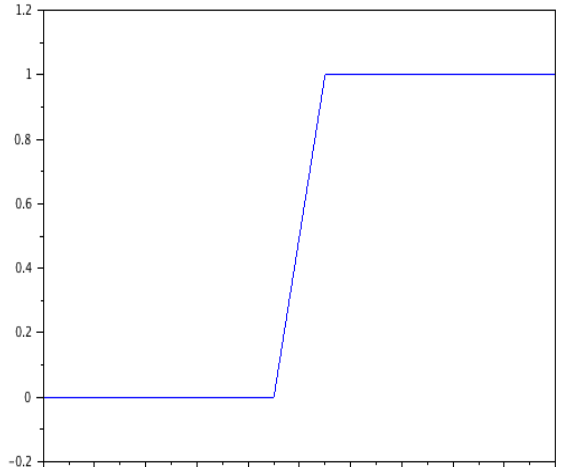
- Idée : contour = **variation rapide** de l'intensité
  - Comment le détecter ?

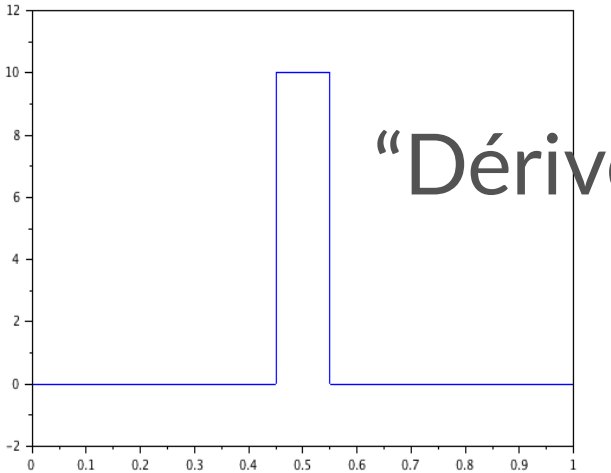


Vers la détection de contours

- Idée : contour = **variation rapide** de l'intensité
  - Comment le détecter ?



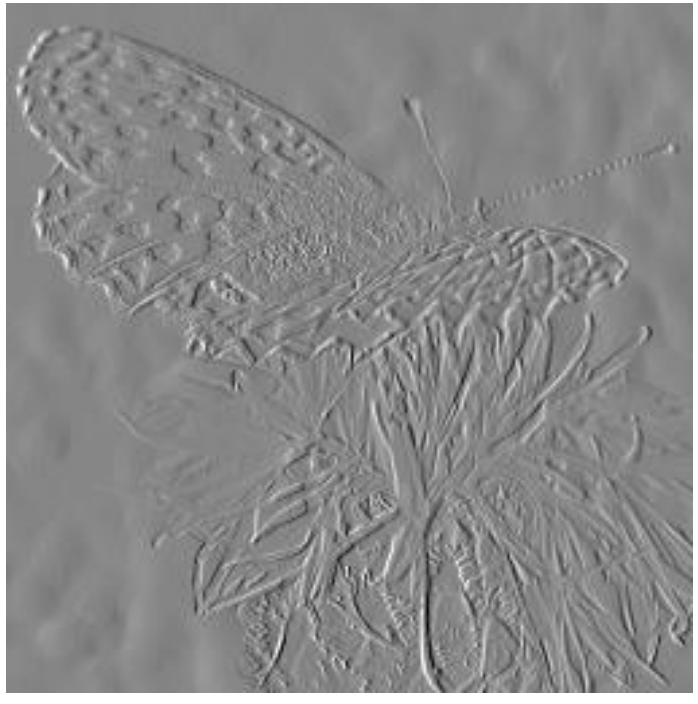




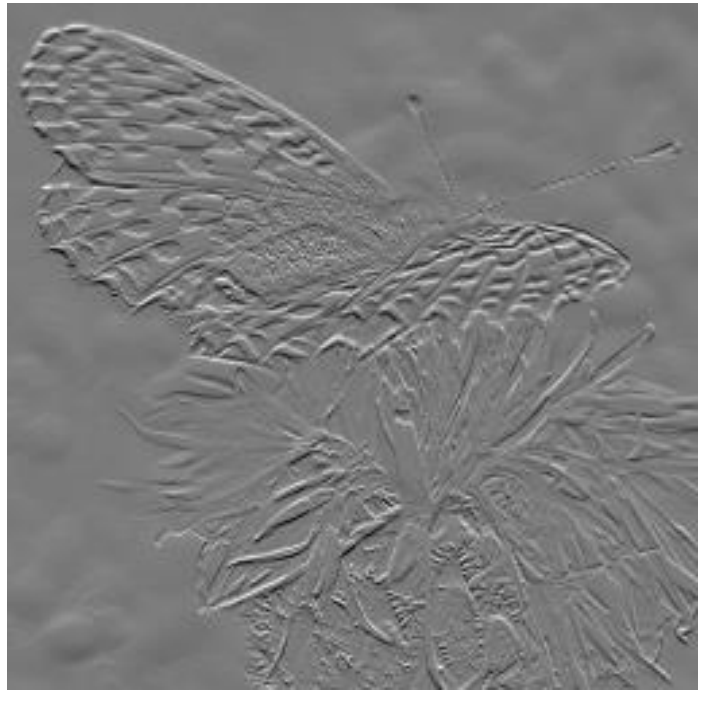
"Dérivée"



$\frac{\partial I}{\partial x}$

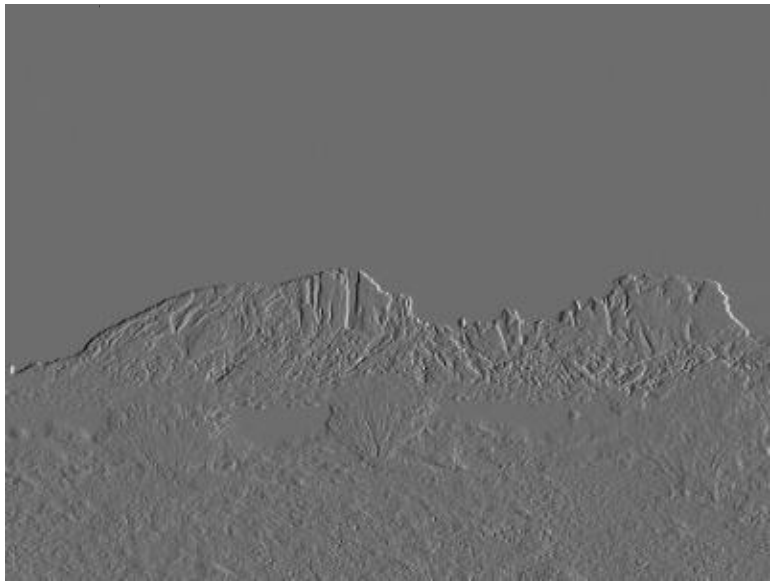


$\frac{\partial I}{\partial y}$

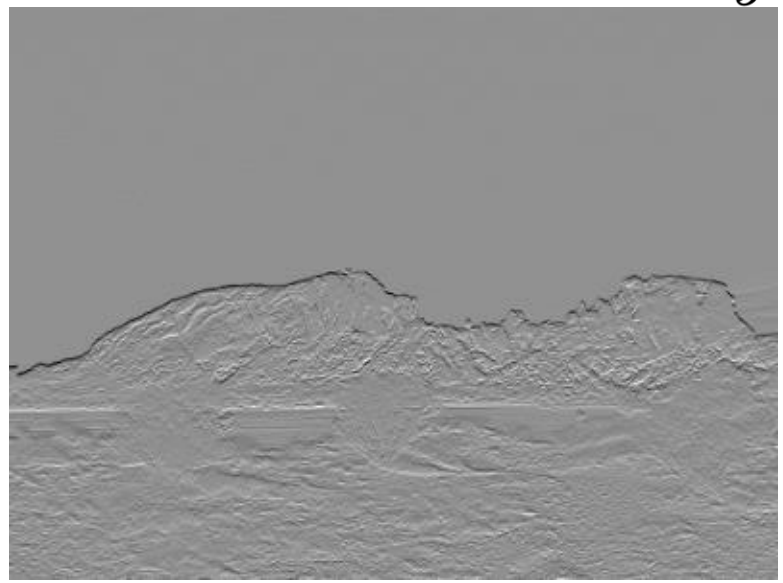




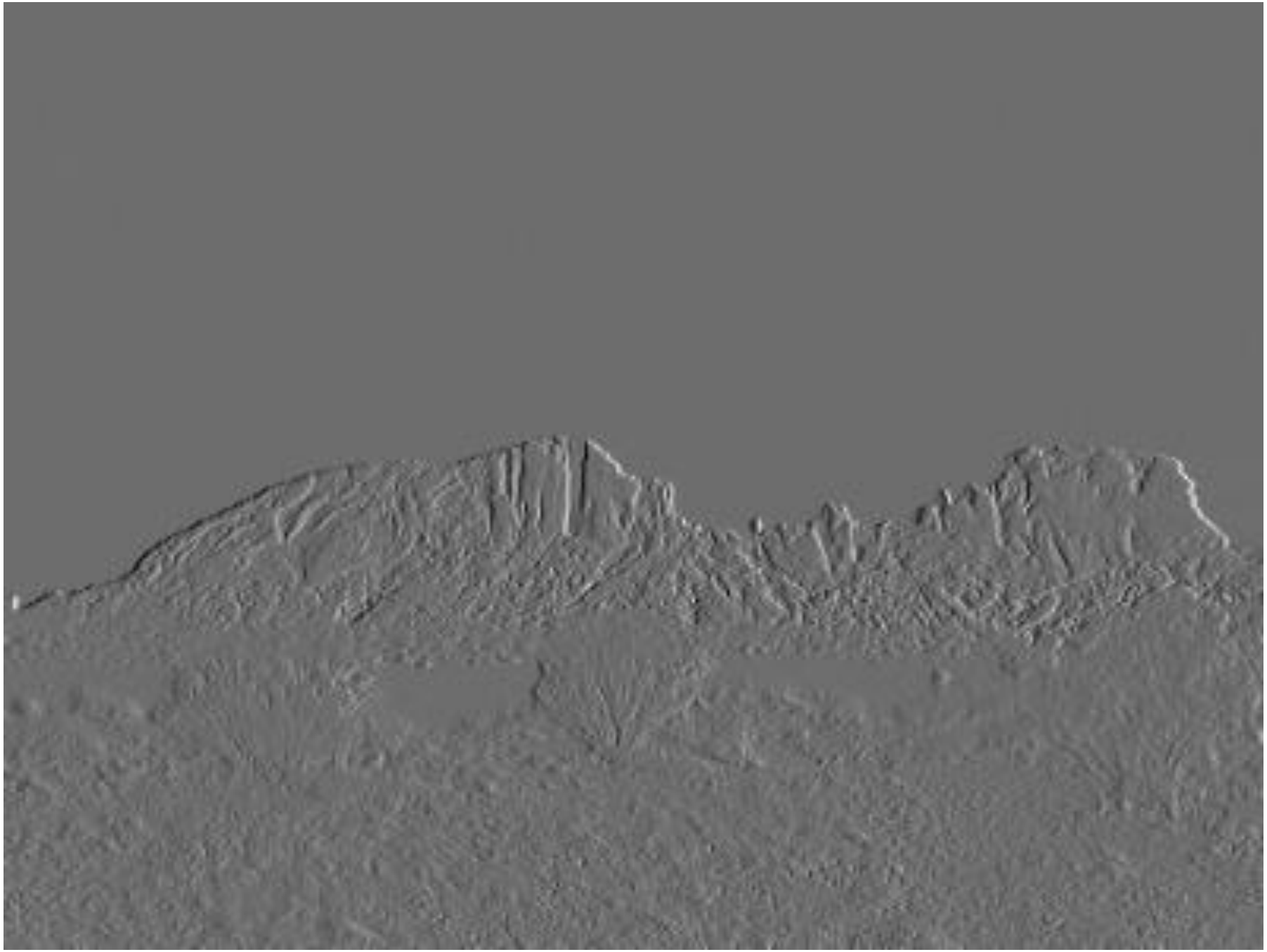
$\frac{\partial I}{\partial x}$

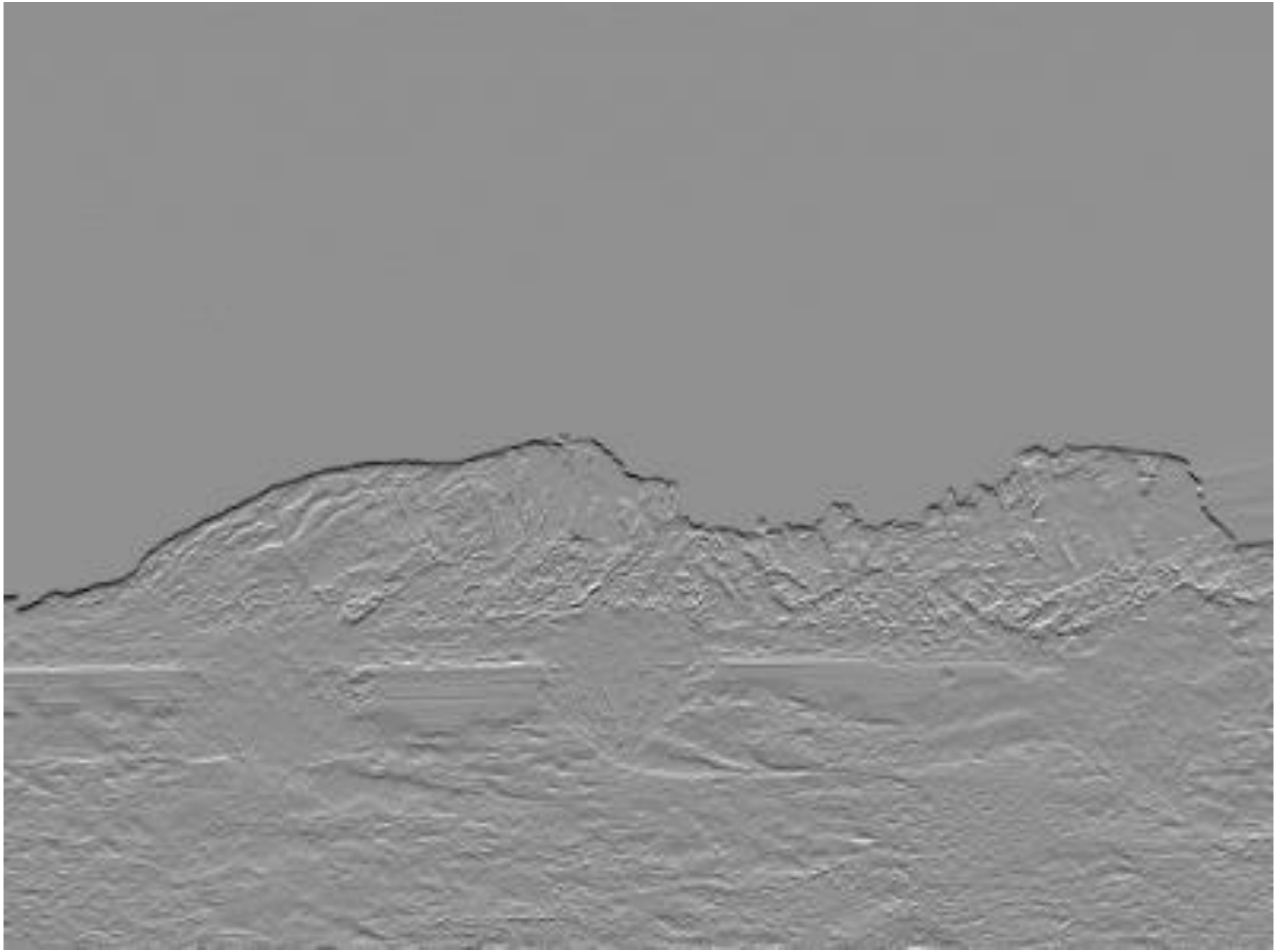


$\frac{\partial I}{\partial y}$



$$\frac{\partial I}{\partial x}$$



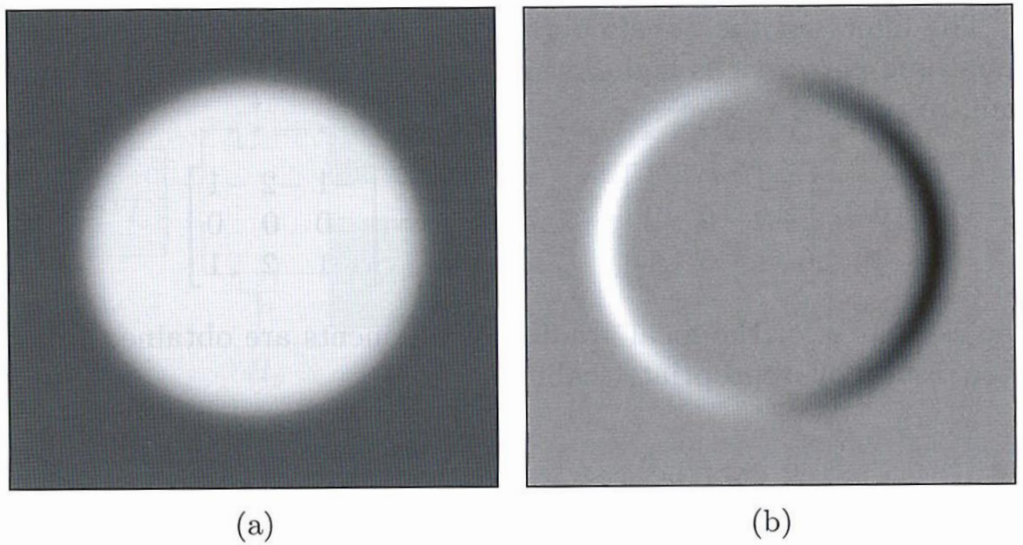


$$\frac{\partial I}{\partial y}$$

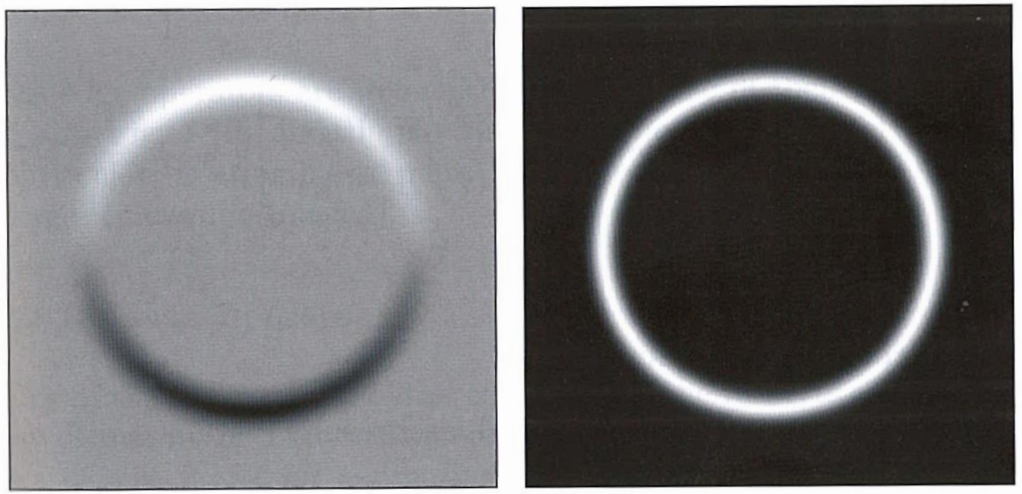
Gradient d'une image continue

Image continue

$I(x, y)$

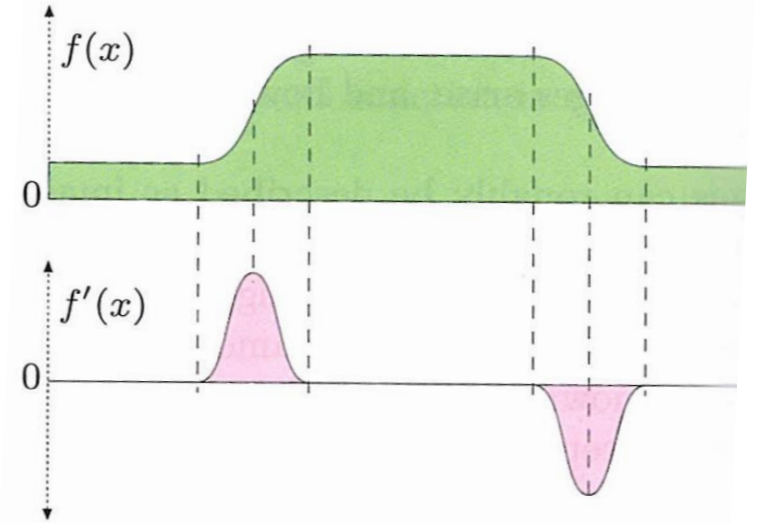


(a) (b)



$\frac{\partial I}{\partial y}(x, y)$ $\|\nabla I\|$

$\frac{\partial I}{\partial x}(x, y)$

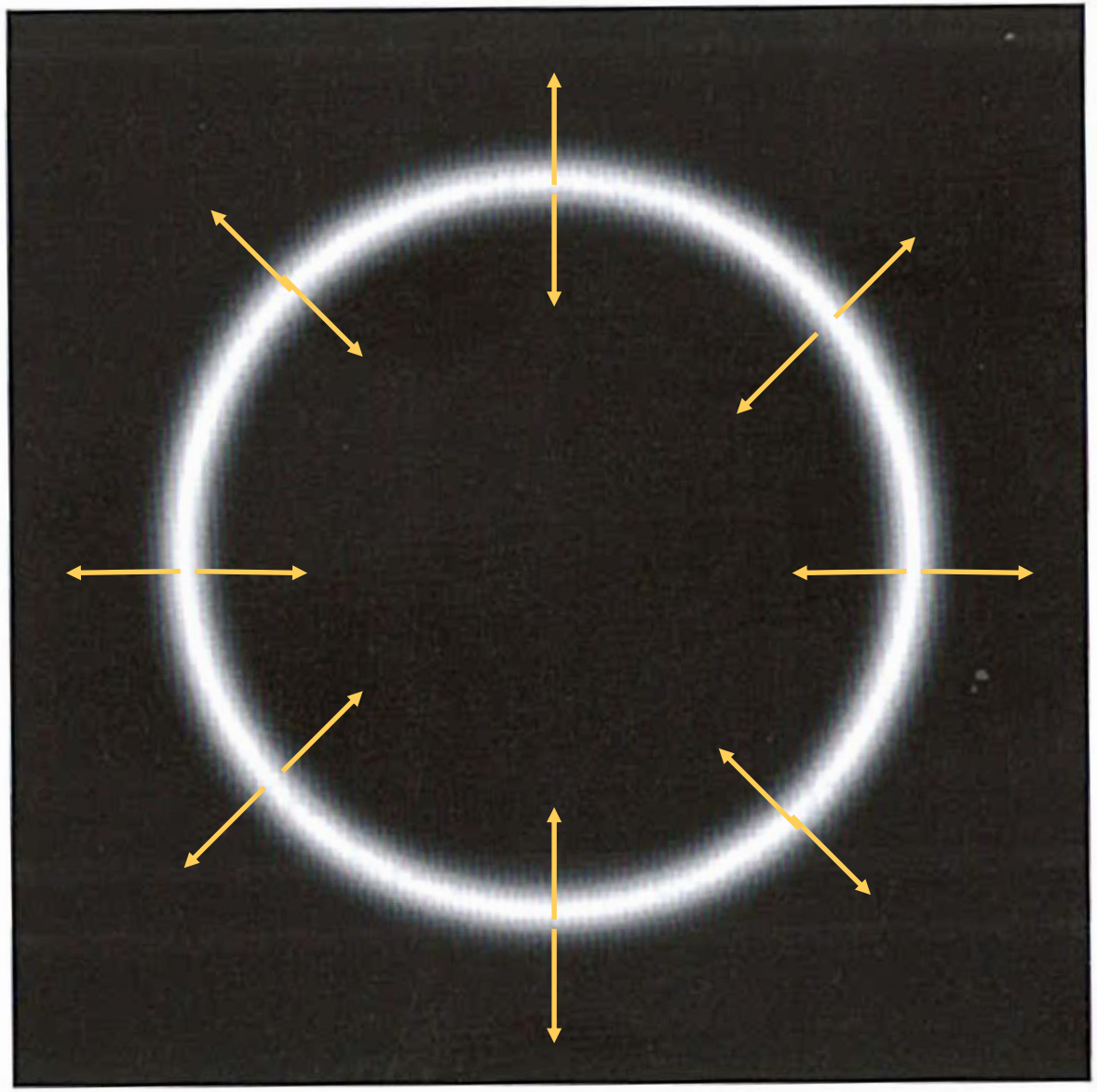


Vecteur gradient :

$$\nabla I = \begin{pmatrix} \frac{\partial I}{\partial x} \\ \frac{\partial I}{\partial y} \end{pmatrix}$$

Norme : $\|\nabla I\| = \sqrt{\left(\frac{\partial I}{\partial x}\right)^2 + \left(\frac{\partial I}{\partial y}\right)^2}$

Gradient d'une image

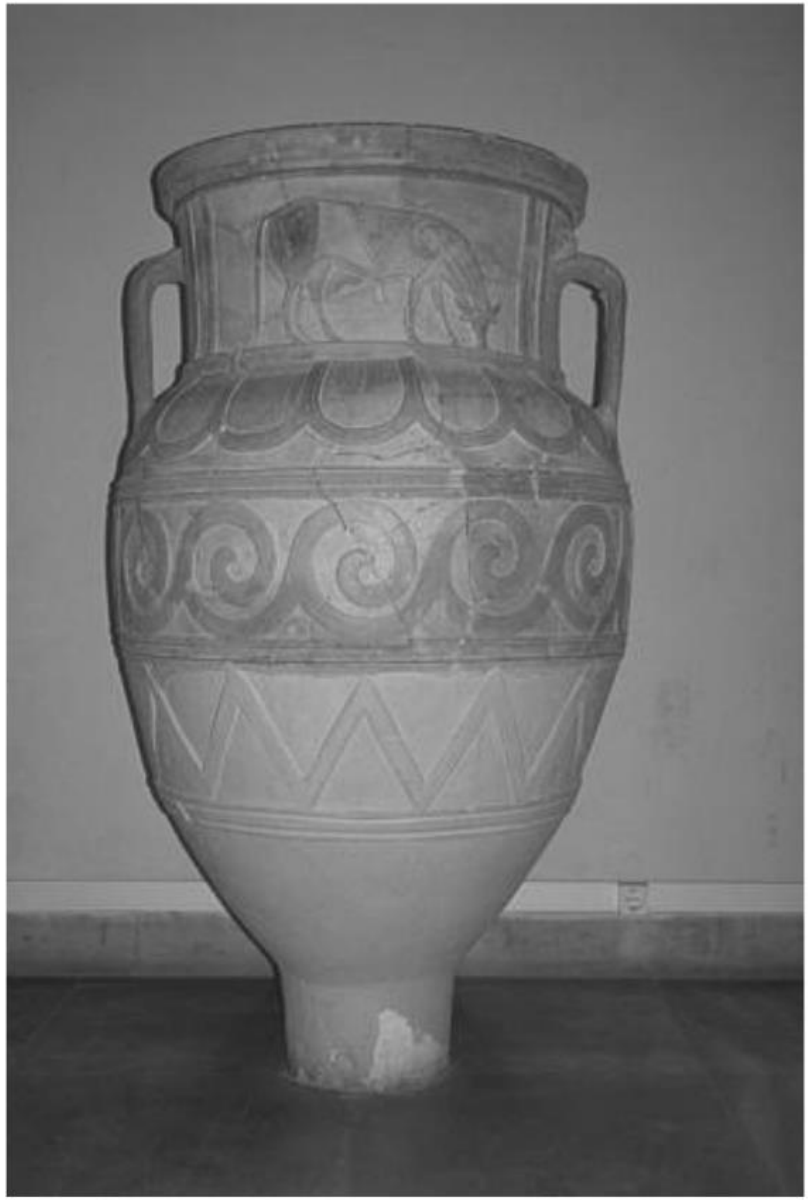


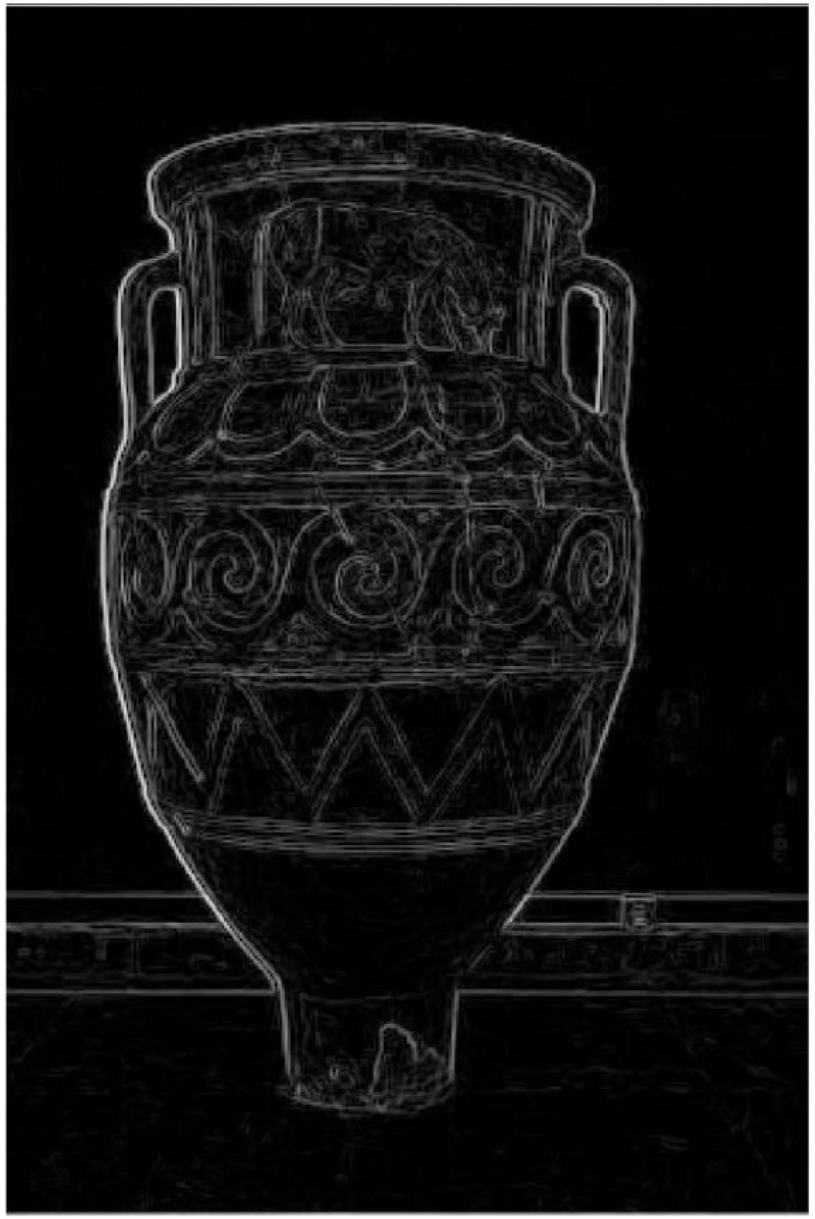
Vecteur gradient :

$$\nabla I = \begin{pmatrix} \frac{\partial I}{\partial x} \\ \frac{\partial I}{\partial y} \end{pmatrix}$$

Norme : $\|\nabla I\| = \sqrt{\left(\frac{\partial I}{\partial x}\right)^2 + \left(\frac{\partial I}{\partial y}\right)^2}$

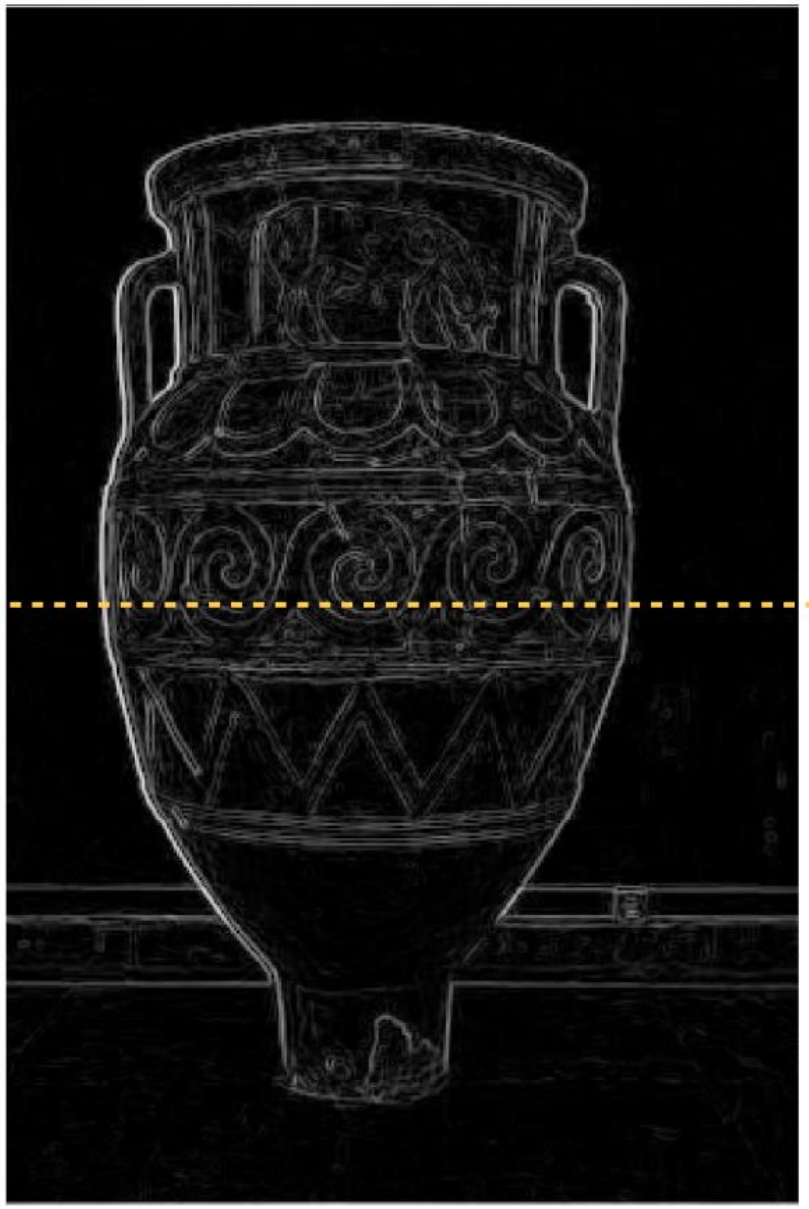
Norme du gradient obtenu par différences finies



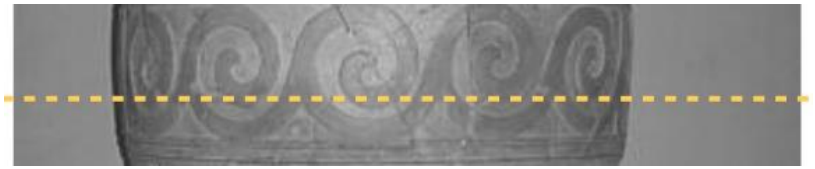


Norme du gradient obtenu par différences finies

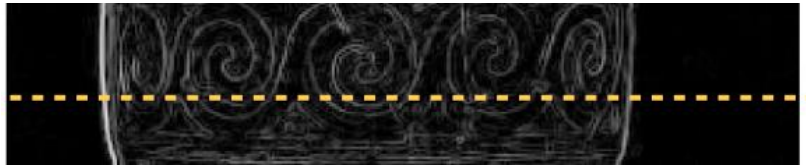




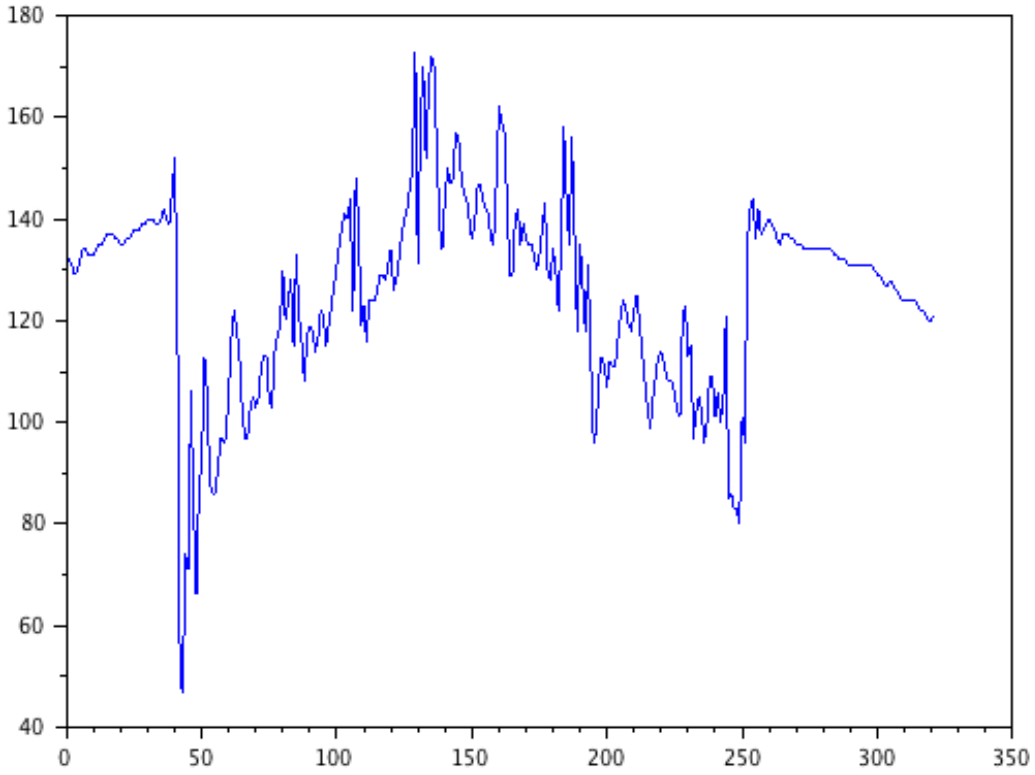
Norme du gradient obtenu par différences finies

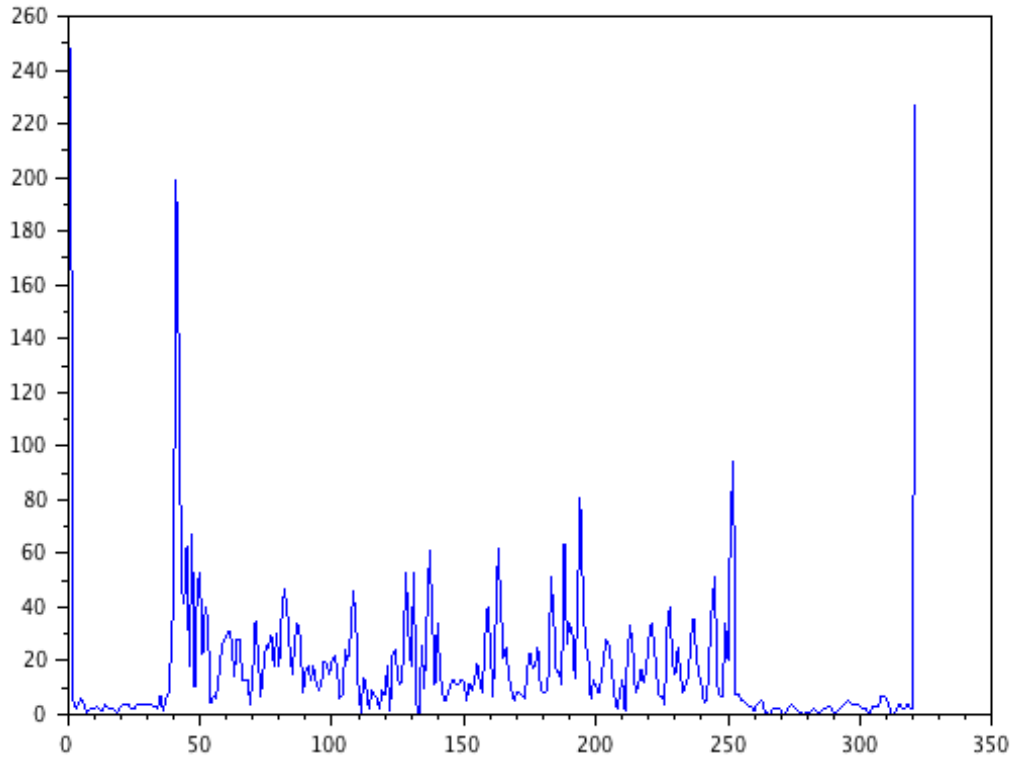


Coupe de l'image



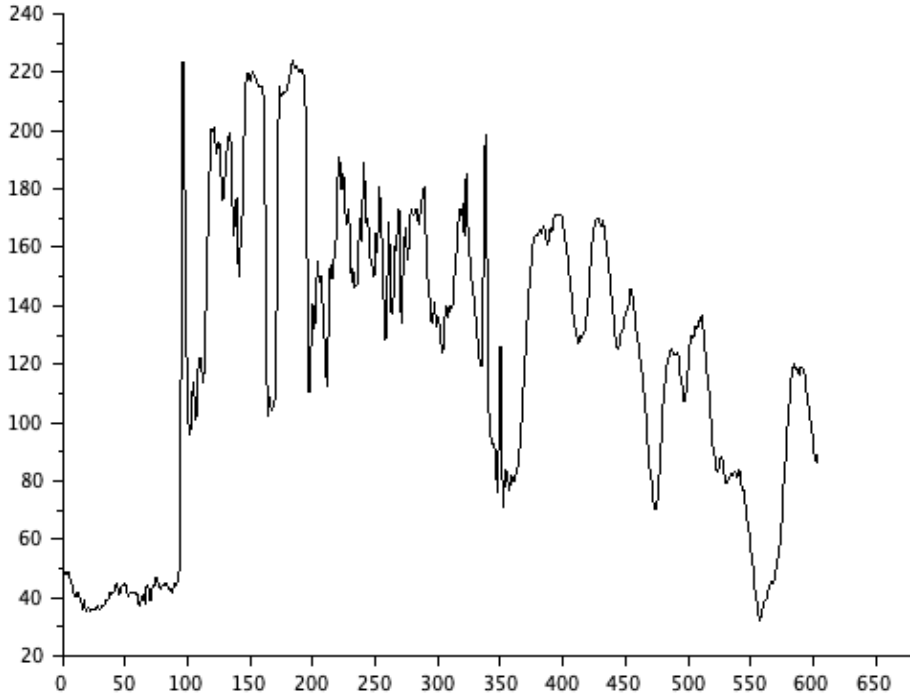
Norme du gradient

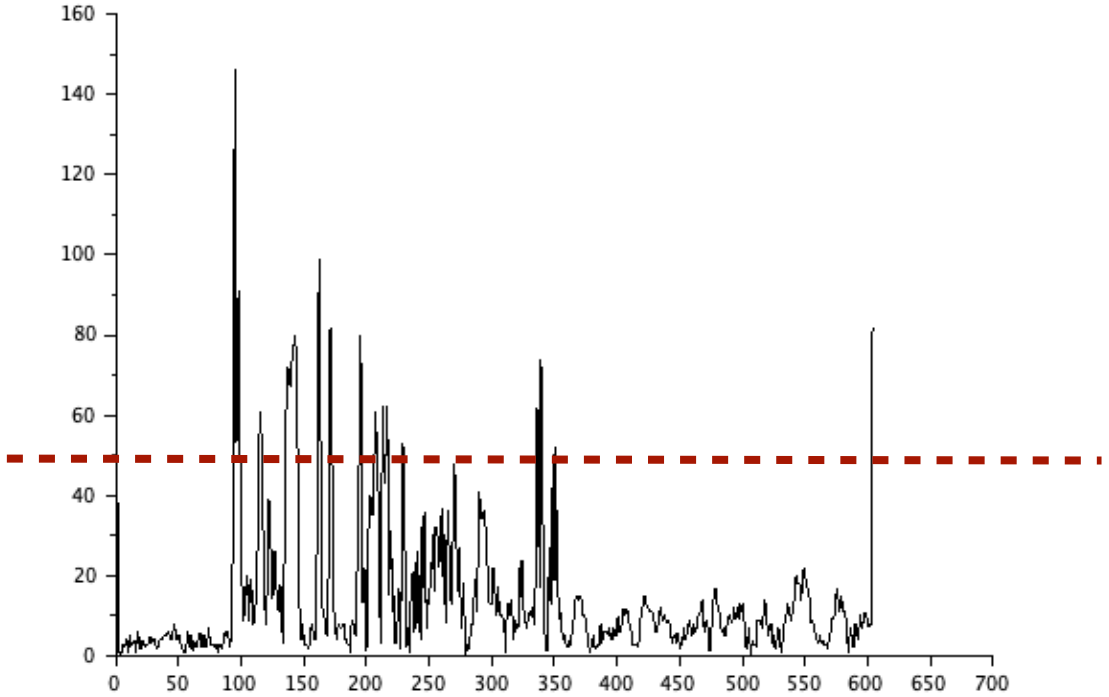




Détection de contours : 1ère méthode

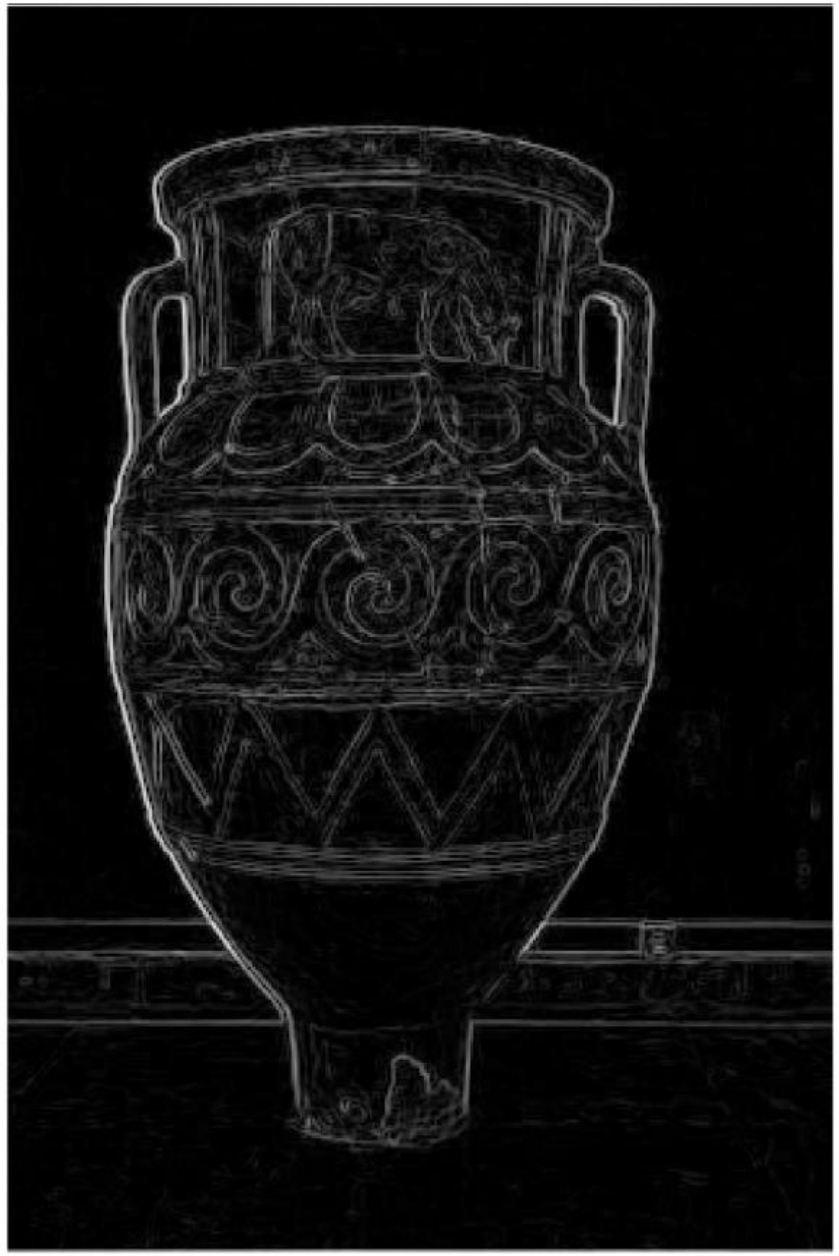
- Contour = norme gradient > seuil

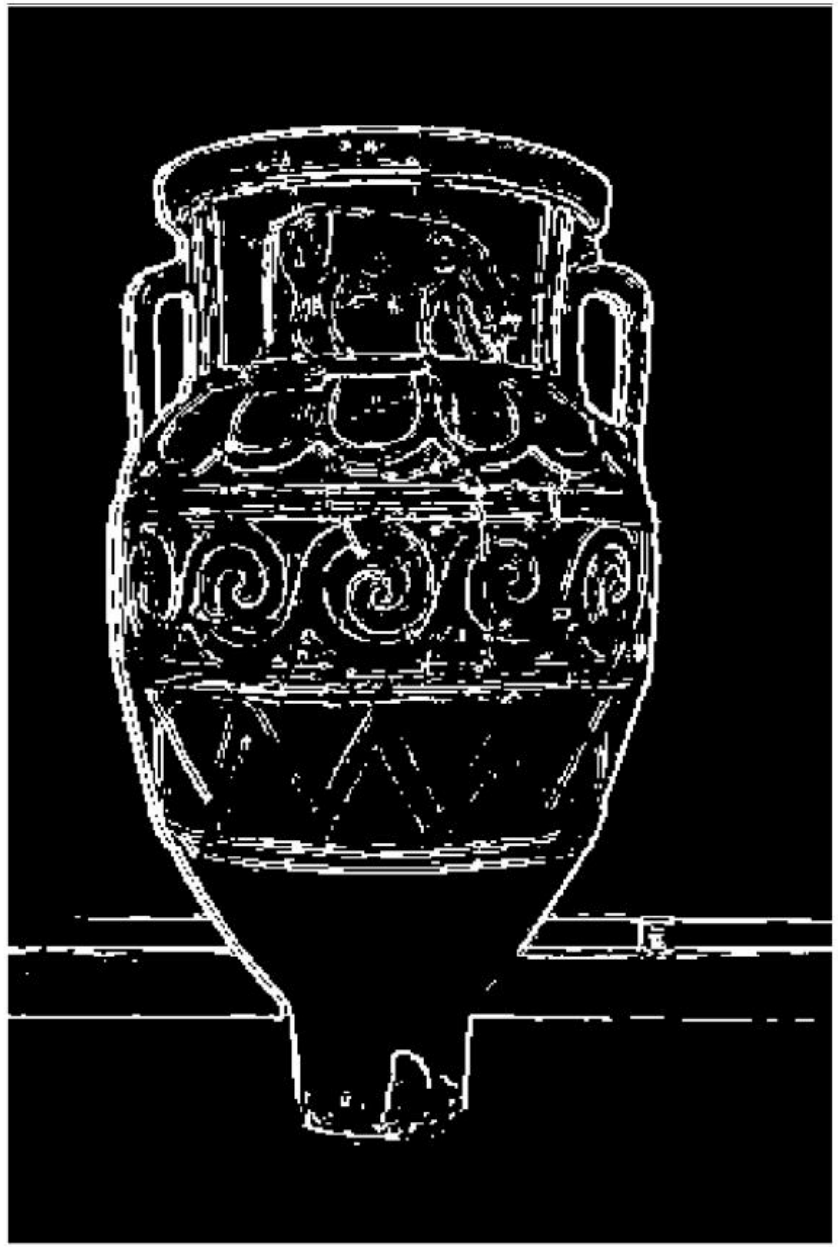




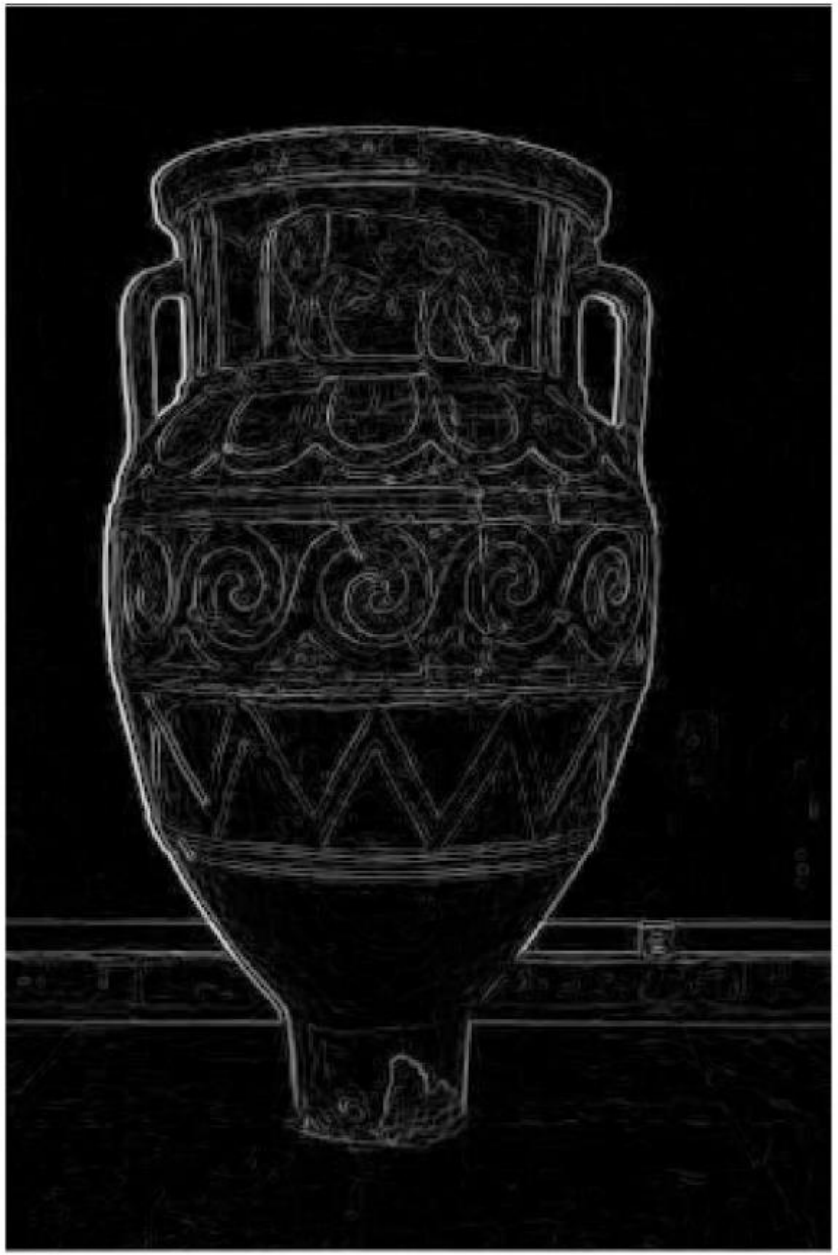
Valeur du seuil ???

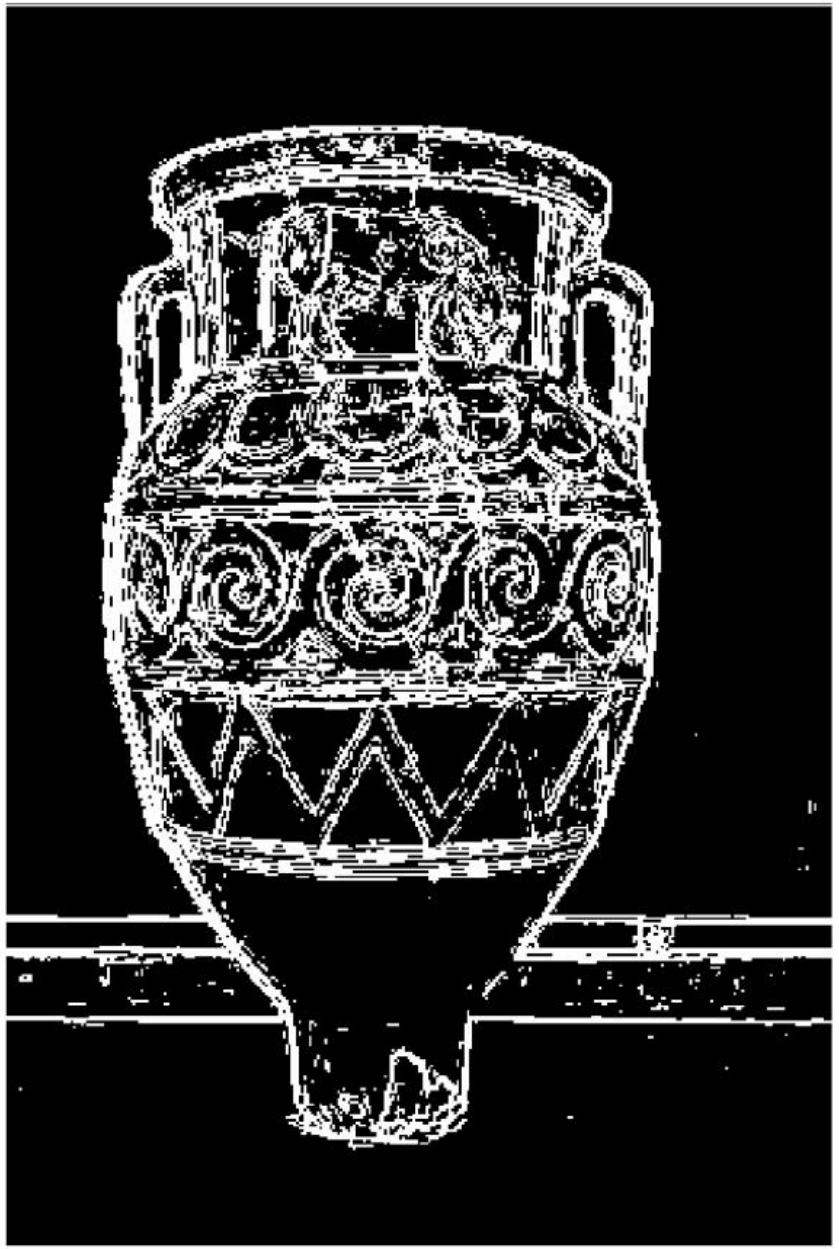
Détection de contours par seuillage (p=90%)



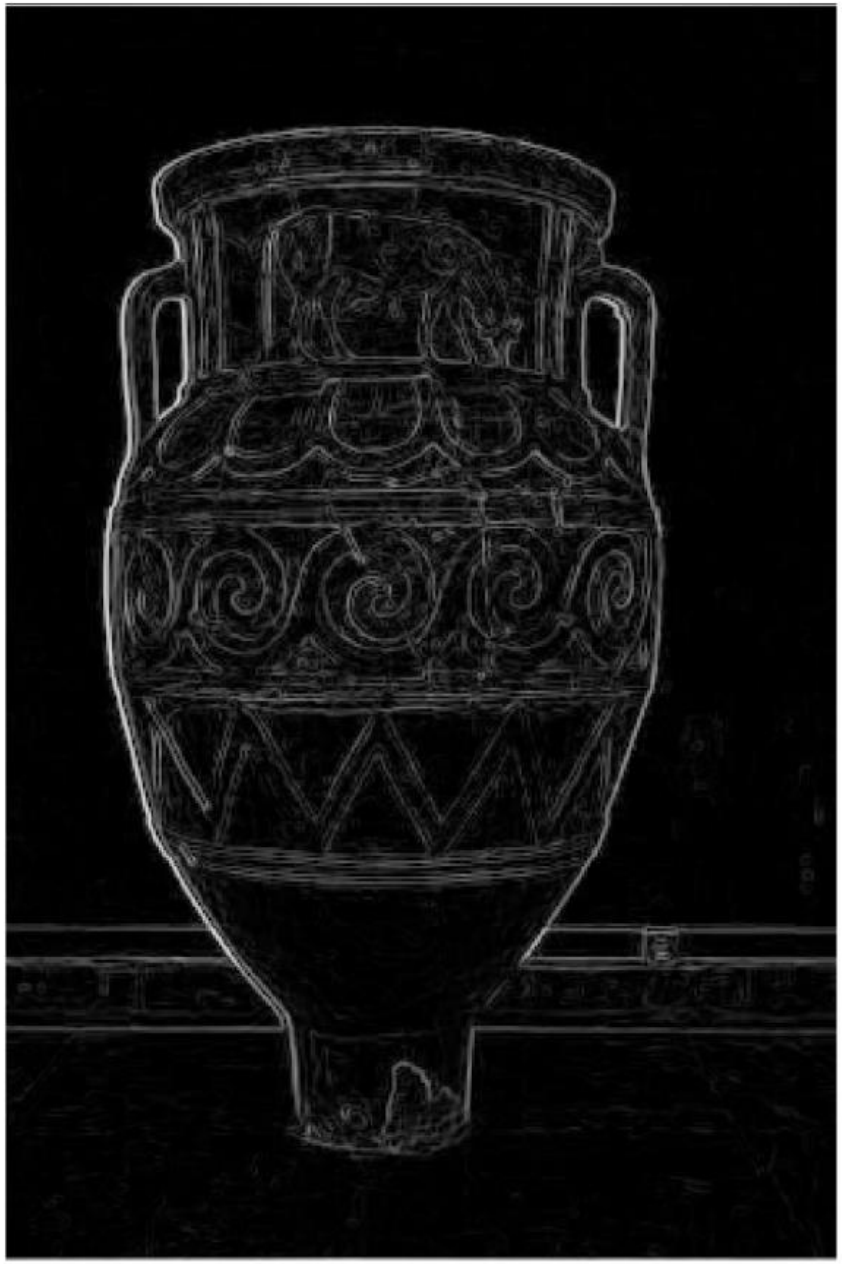


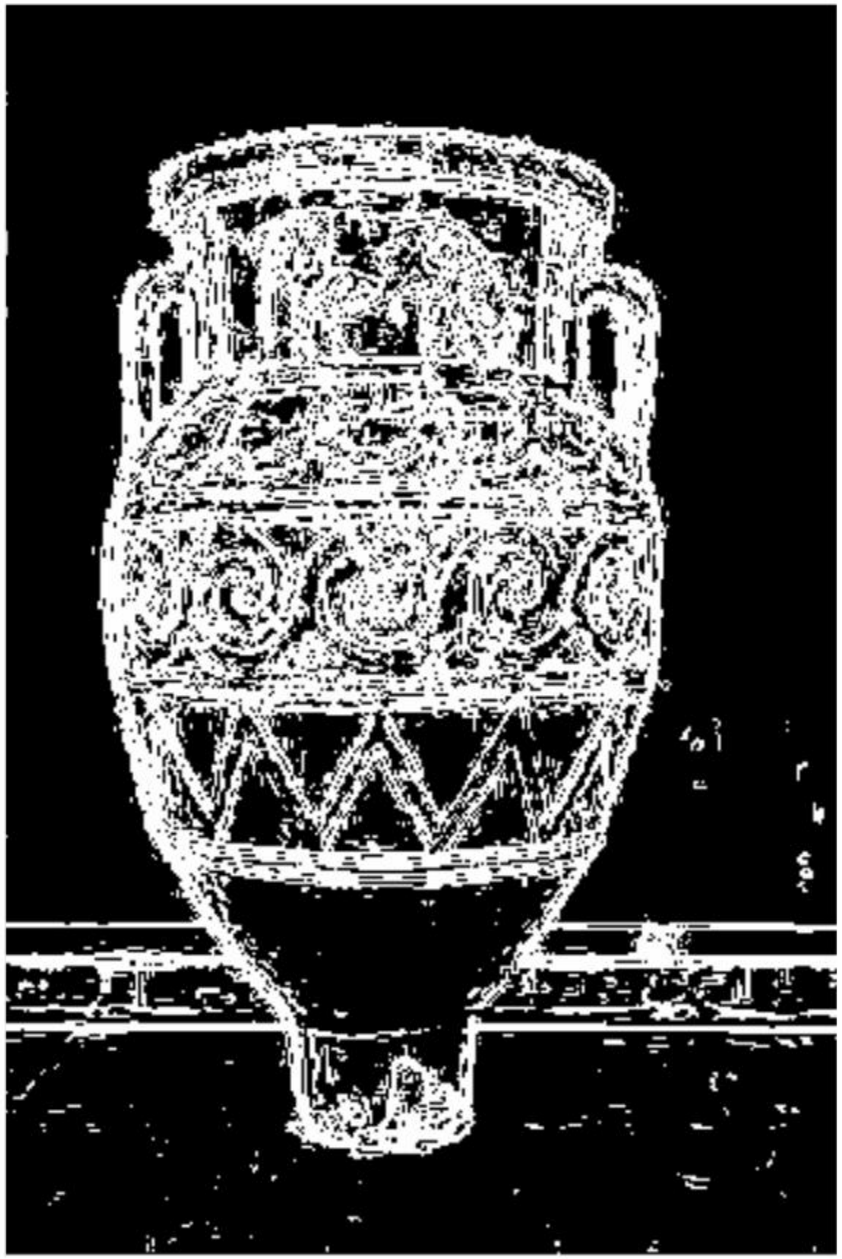
Détection de contours par seuillage (p=80%)





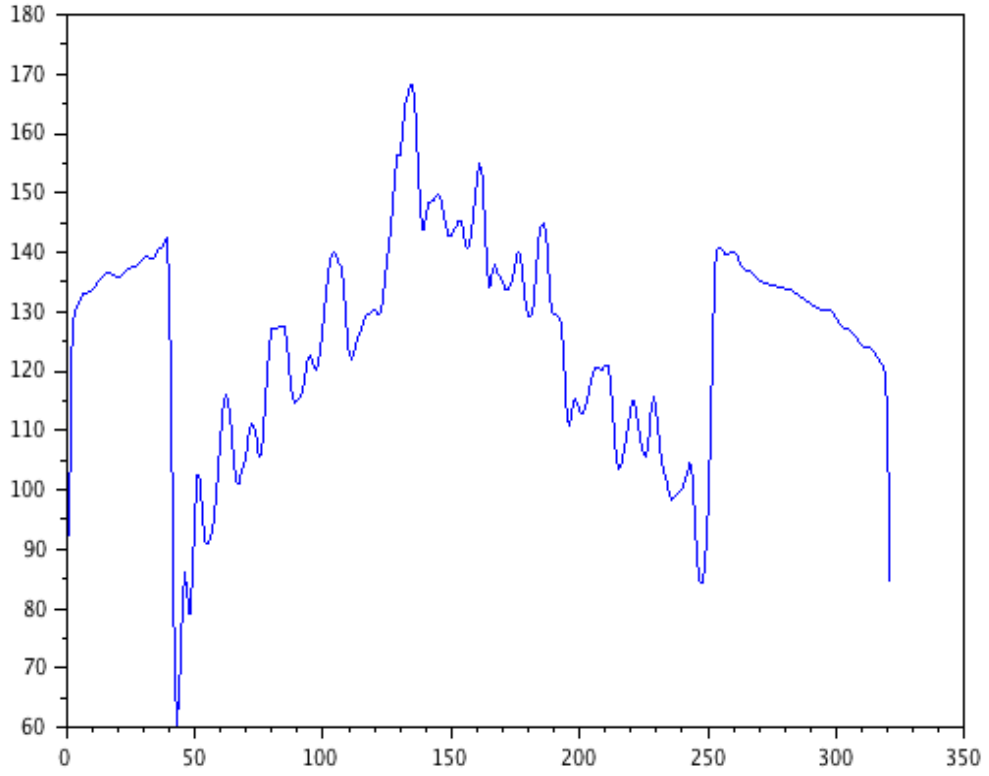
Détection de contours par seuillage (p=70%)

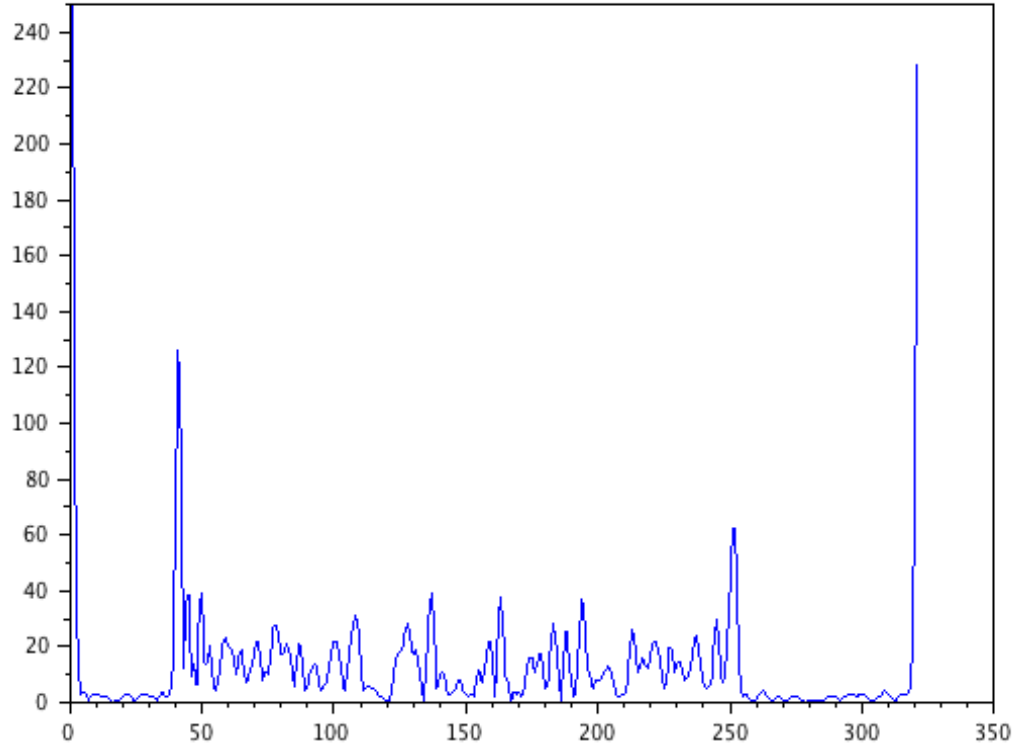




Effet du lissage

- Filtre Gaussien σ = 1

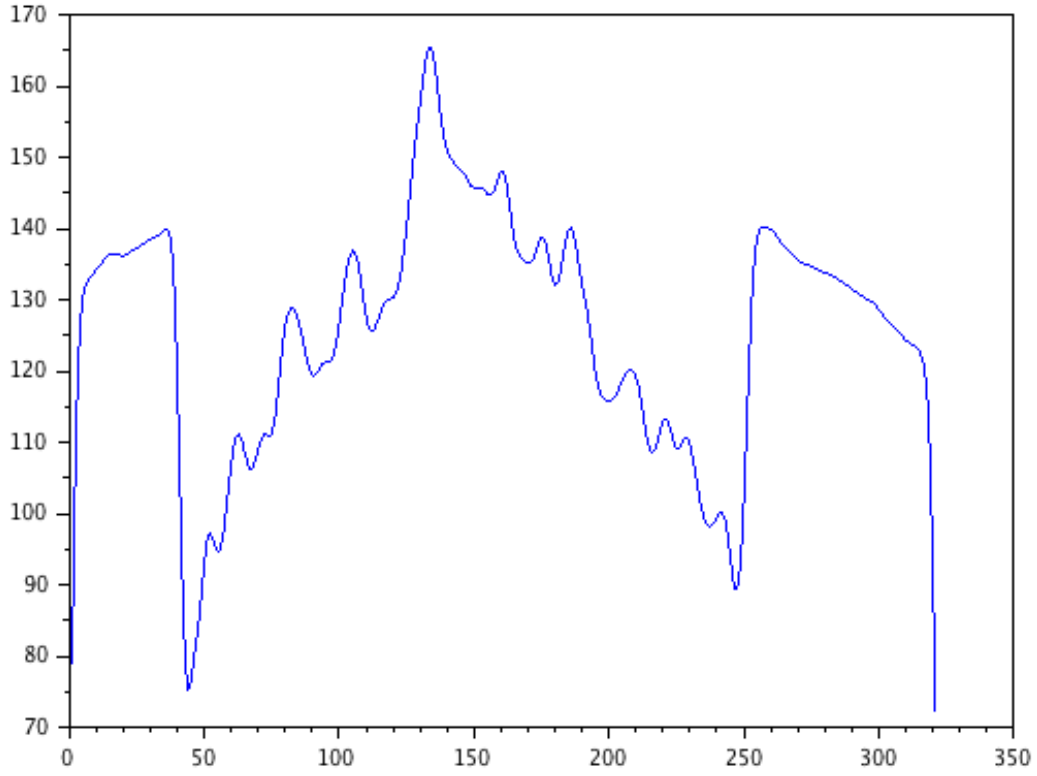


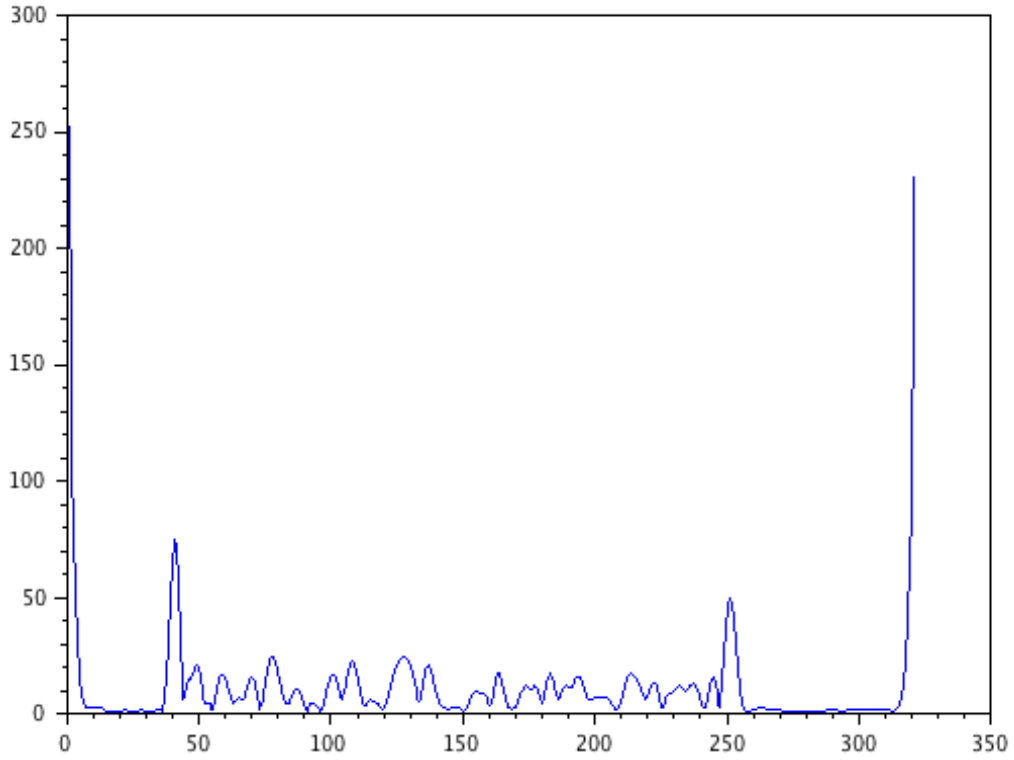


Effet du lissage

- Filtre Gaussien σ = 2

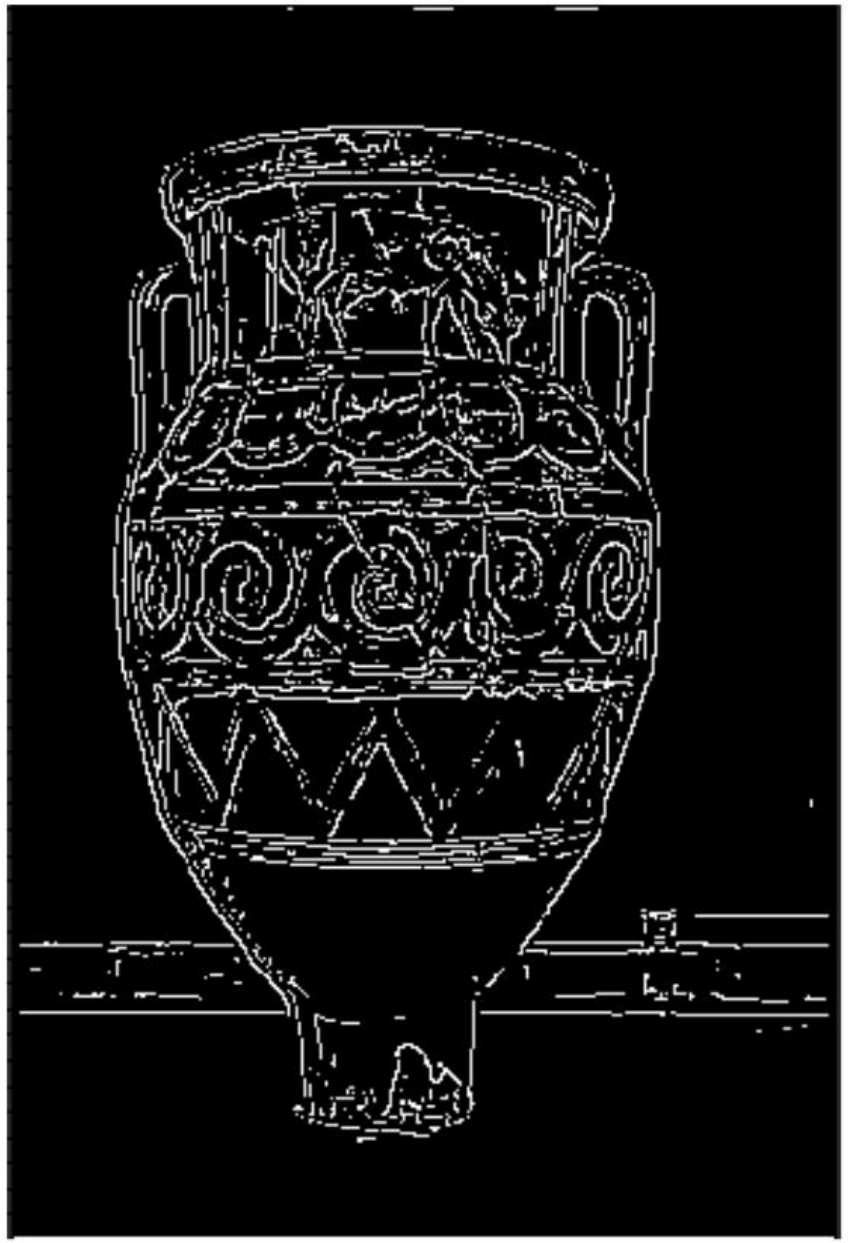
Rend l'image plus "dérivable" grâce à la convolution par la Gaussienne





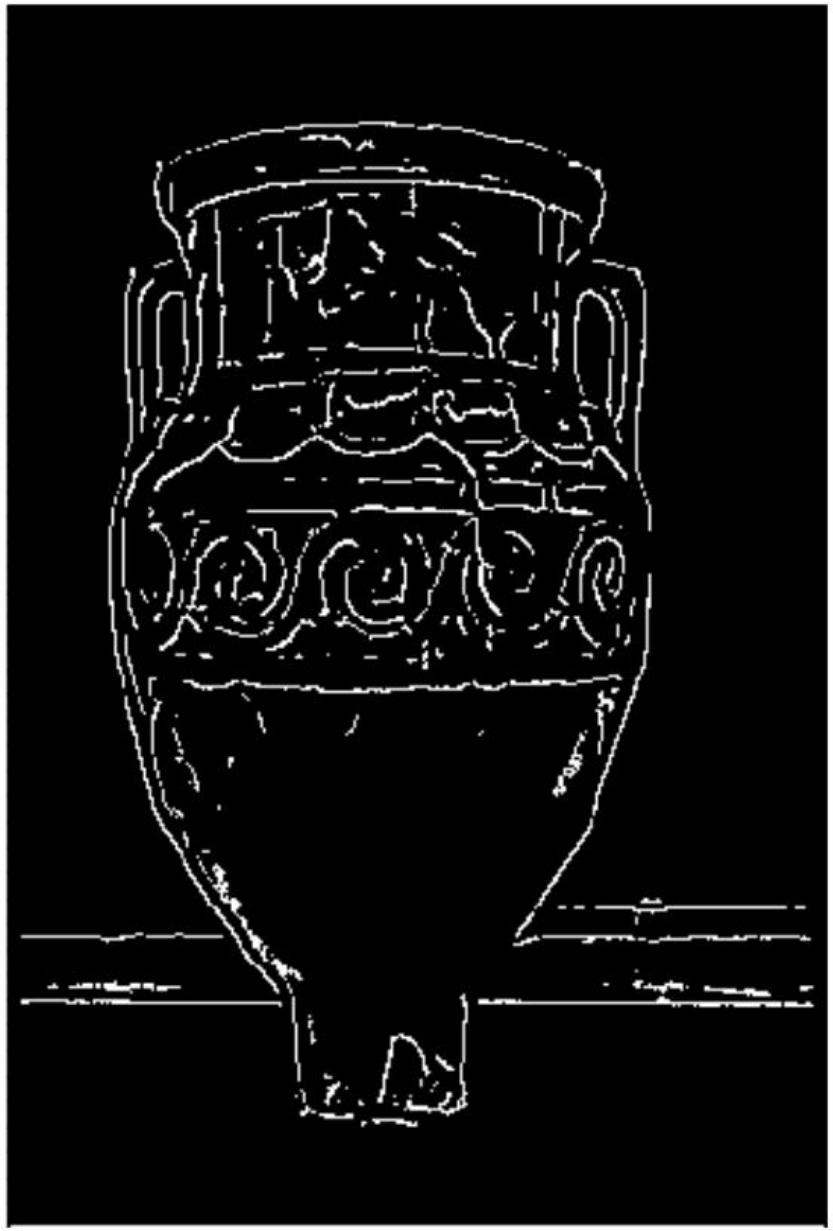
Lissage Gaussien (σ = 1) +Contour





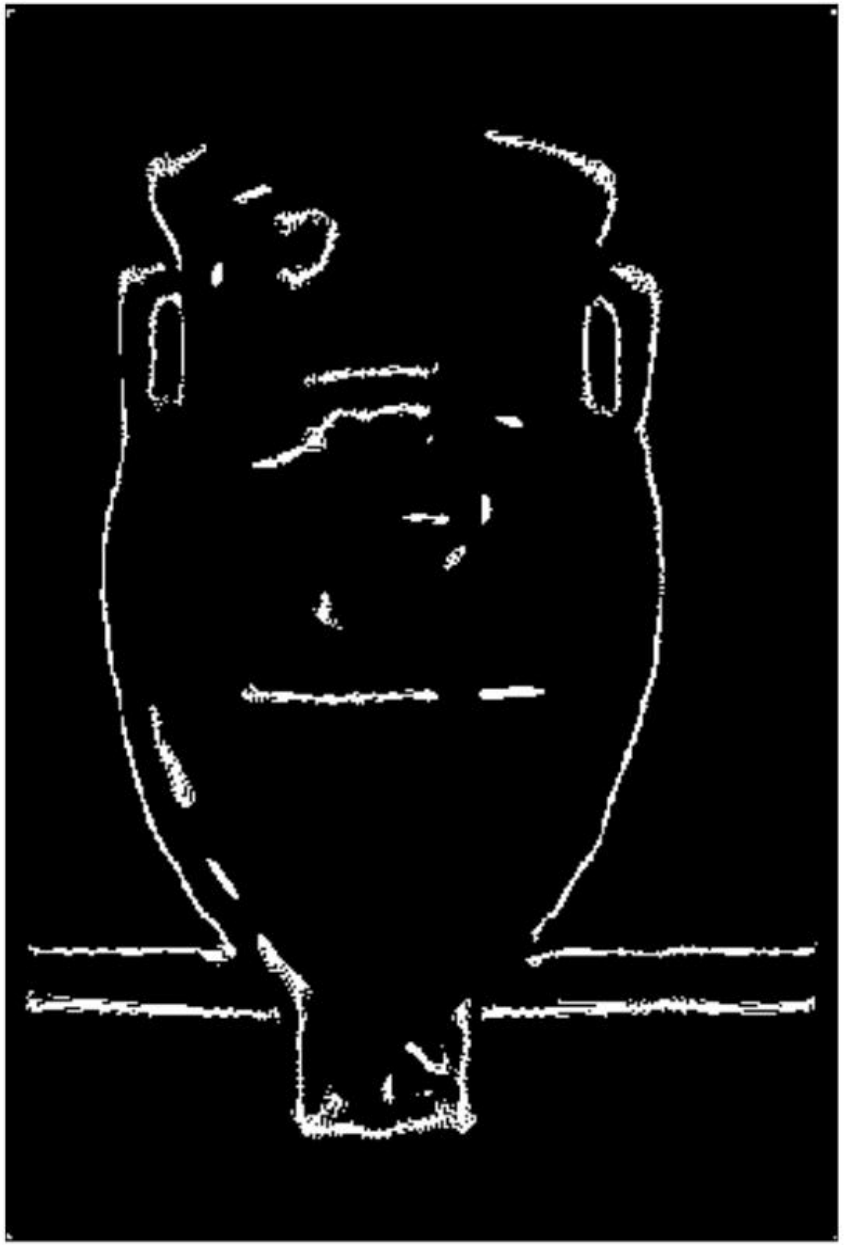
Lissage Gaussien (σ = 2) +Contour



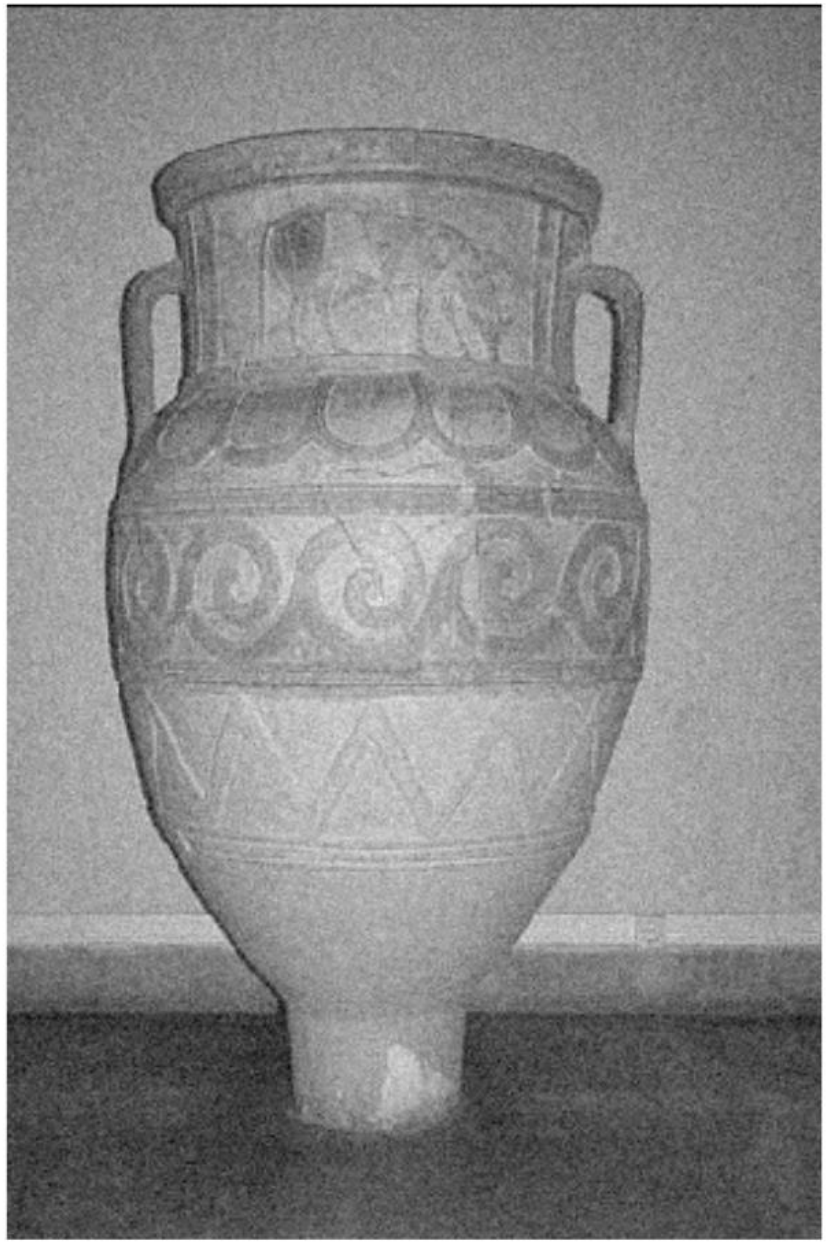


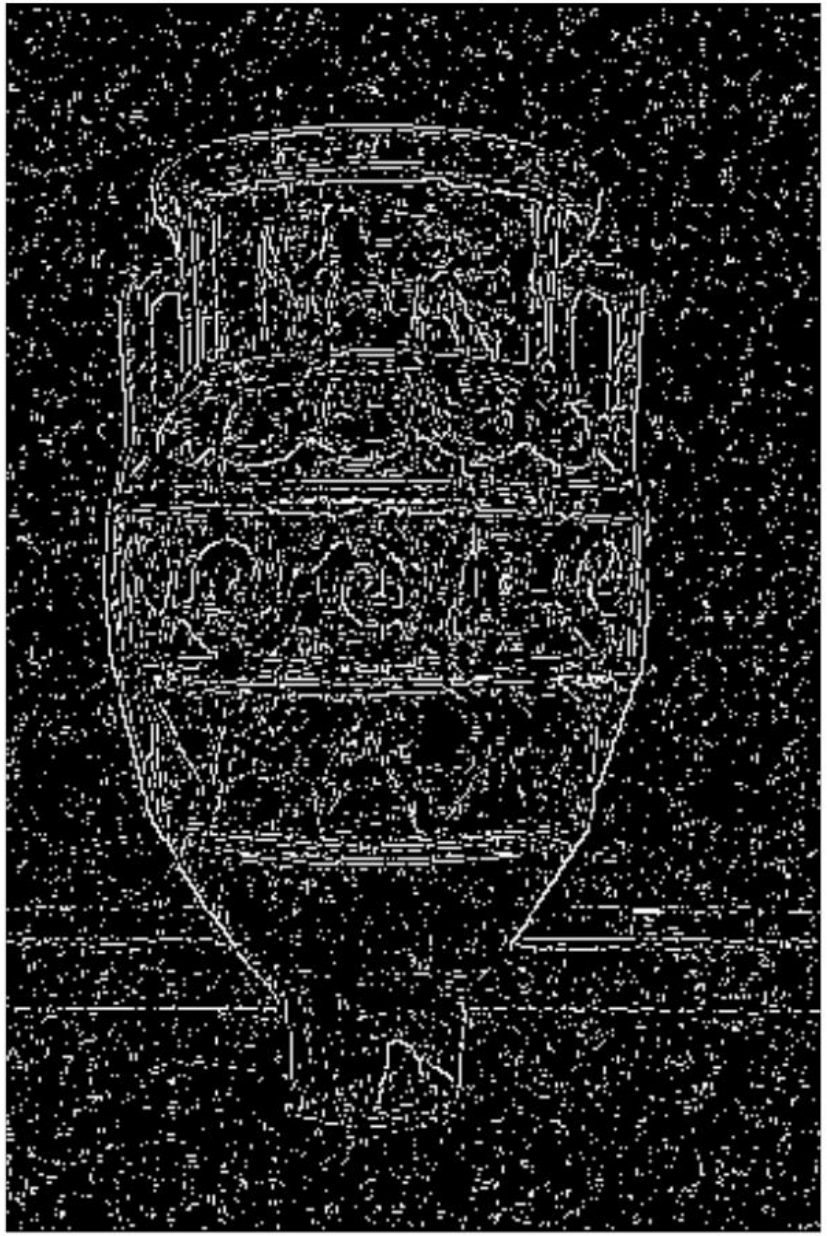
Lissage Gaussien (σ = 5) +Contour



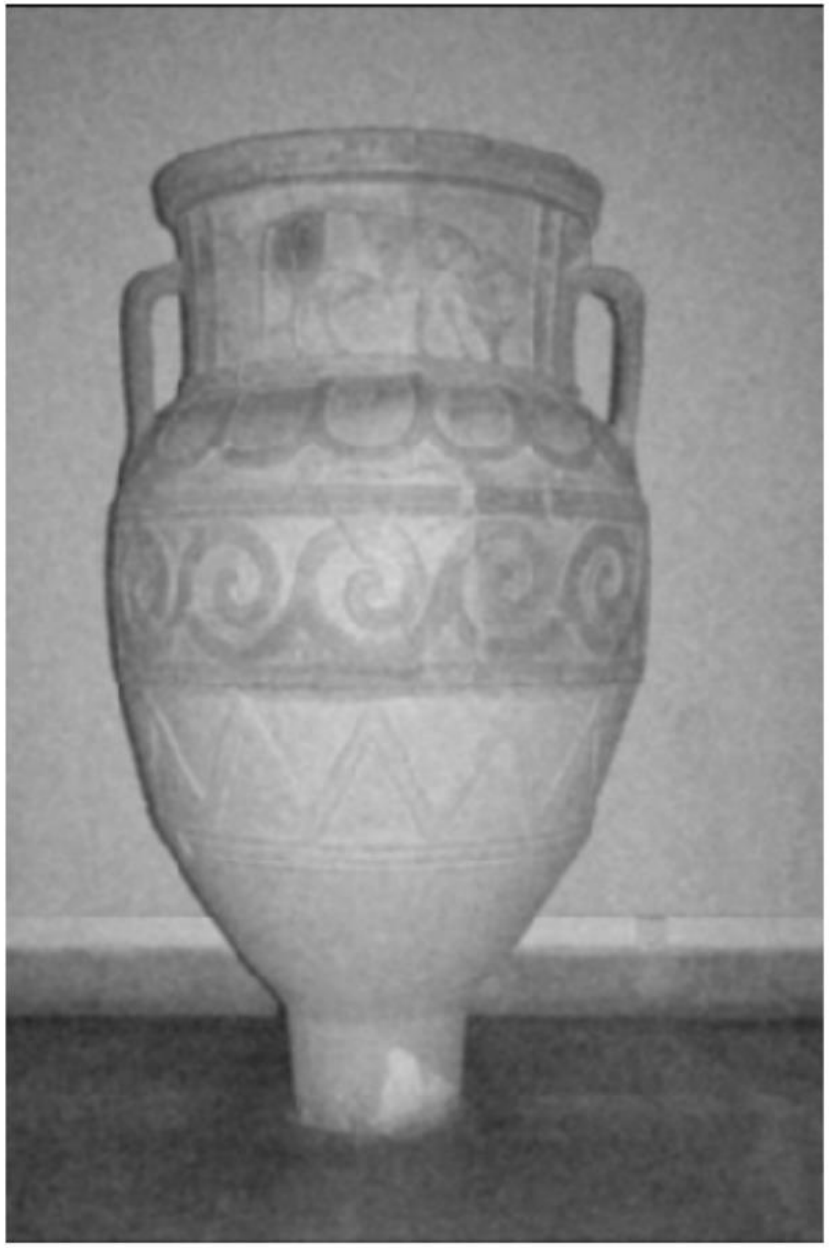


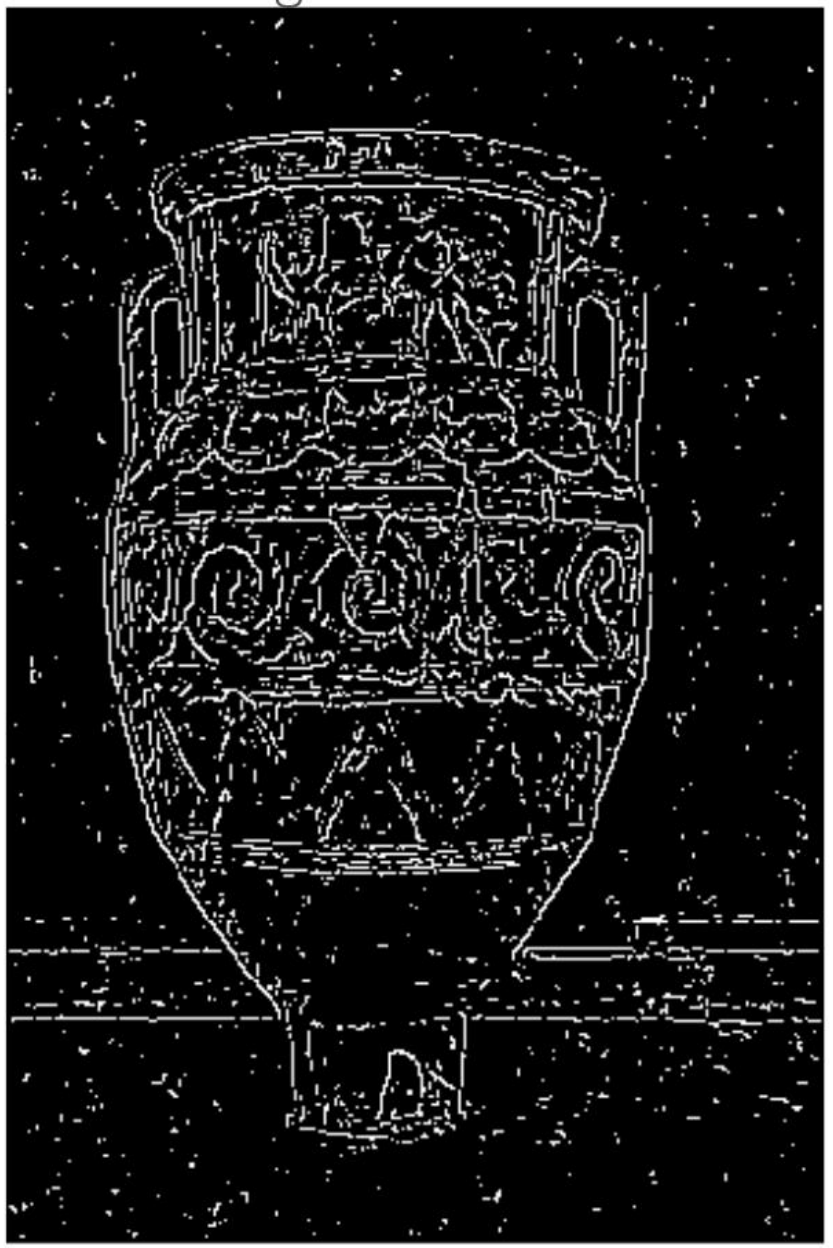
Effet du bruit





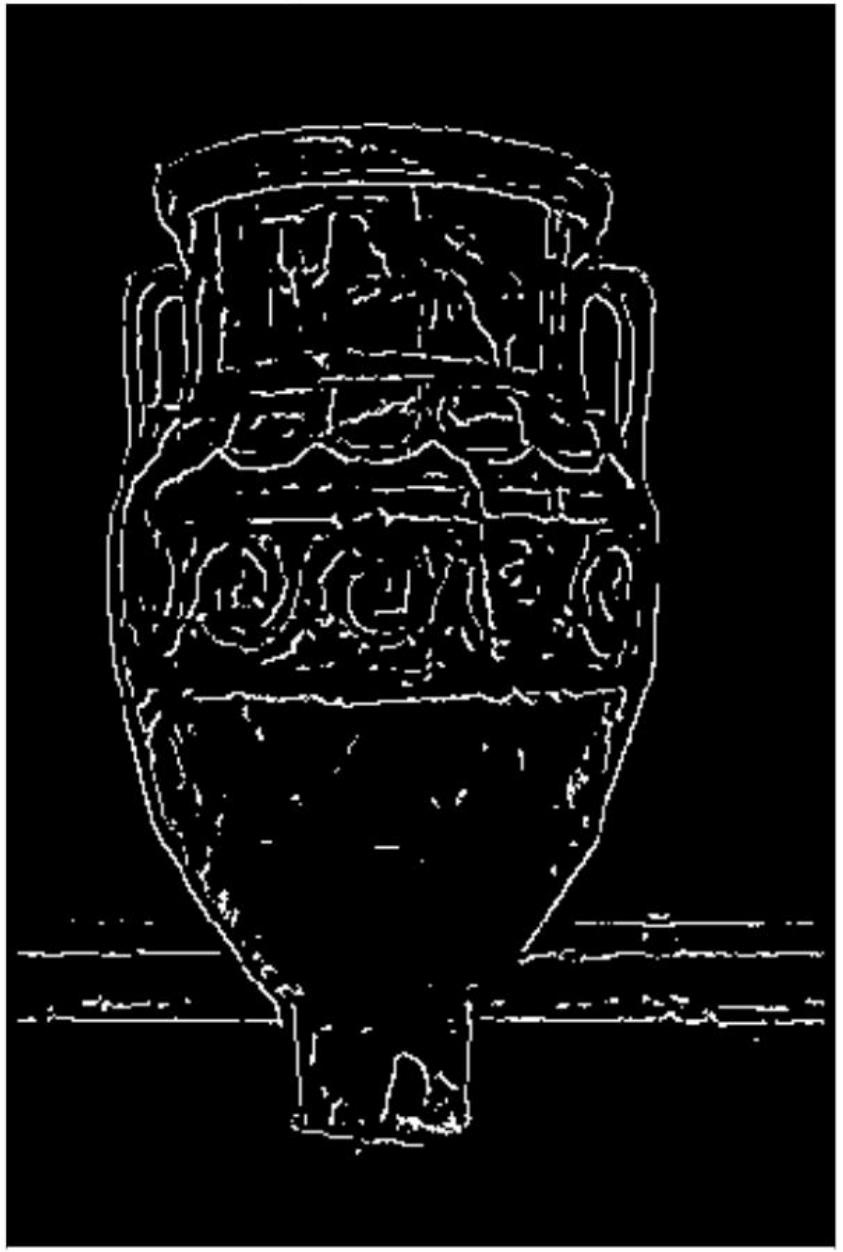
Effet du bruit : Lissage Gaussien (σ = 1) +Contour





Effet du bruit : Lissage Gaussien (σ = 2) +Contour





Exercices récapitulatifs

Ex. 1

Le filtre $\begin{matrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{matrix}$ est il un filtre de moyennage local, de gradient en x (horizontal), de gradient en y (vertical) ? Justifiez votre réponse.

Appliquez le filtre de la question 2) à l'image ci-dessous.

```
0.  0.  0.  0.  0.  0.
0.  0.  0.  0.  0.  0.
0.  0.  0.  0.  0.  0.
0.  0.  1.  1.  1.  1.
0.  0.  1.  1.  1.  1.
0.  0.  1.  1.  1.  1.
```

Calculez la norme du gradient de l'image ci-dessus.

Ex. 2





On peut remarquer que les moustaches sont dédoublées sur l'image du contour ; plus généralement, c'est le cas pour les traits (par opposition aux bords des objets pleins). L'objectif de cet exercice est de comprendre pourquoi. Pour simplifier l'analyse, on s'intéresse à une image I qui contient un trait noir vertical sur fond blanc, codée en niveaux de gris sur 8 bits :

```
I  =
   255   255   255   255   255   0   255   255   255   255   255
   255   255   255   255   255   0   255   255   255   255   255
   255   255   255   255   255   0   255   255   255   255   255
   255   255   255   255   255   0   255   255   255   255   255
   255   255   255   255   255   0   255   255   255   255   255
```

Ex. 2

3. Appliquer le filtre D_x à l'image I pour calculer une approximation I_x de sa dérivée suivant x. Comme toutes les lignes sont identiques, vous n'indiquerez dans votre réponse qu'une ligne de I_x (et de même pour les images des questions suivantes). Pour le traitement des pixels du bord de l'image, on fait le choix de leur donner la valeur nulle : ainsi la première et la dernière colonne de I_x seront nulles et on n'appliquera la formule précédente que sur les autres colonnes.
4. Calculer l'image I_N correspondant à la norme du gradient de l'image I. L'image I_C contenant